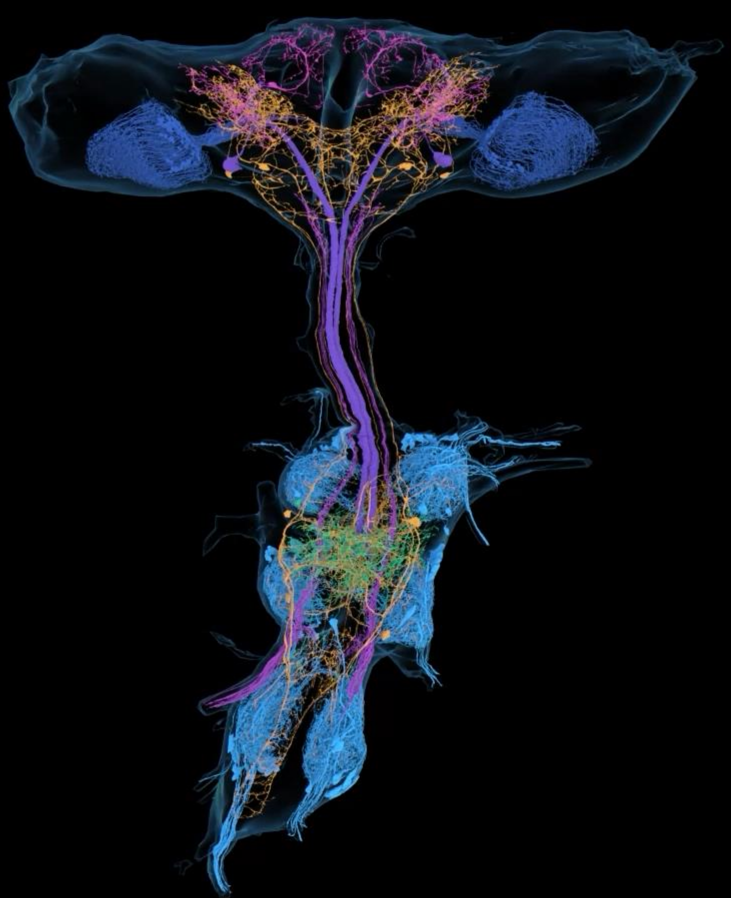
Wei-Chung Lee Lab of
Harvard Medical School
with Murthy & Seung Labs of
Princeton Neuroscience Institute
present



The BANC

Brain and Nerve Cord

of

Drosophila melanogaster

New open access

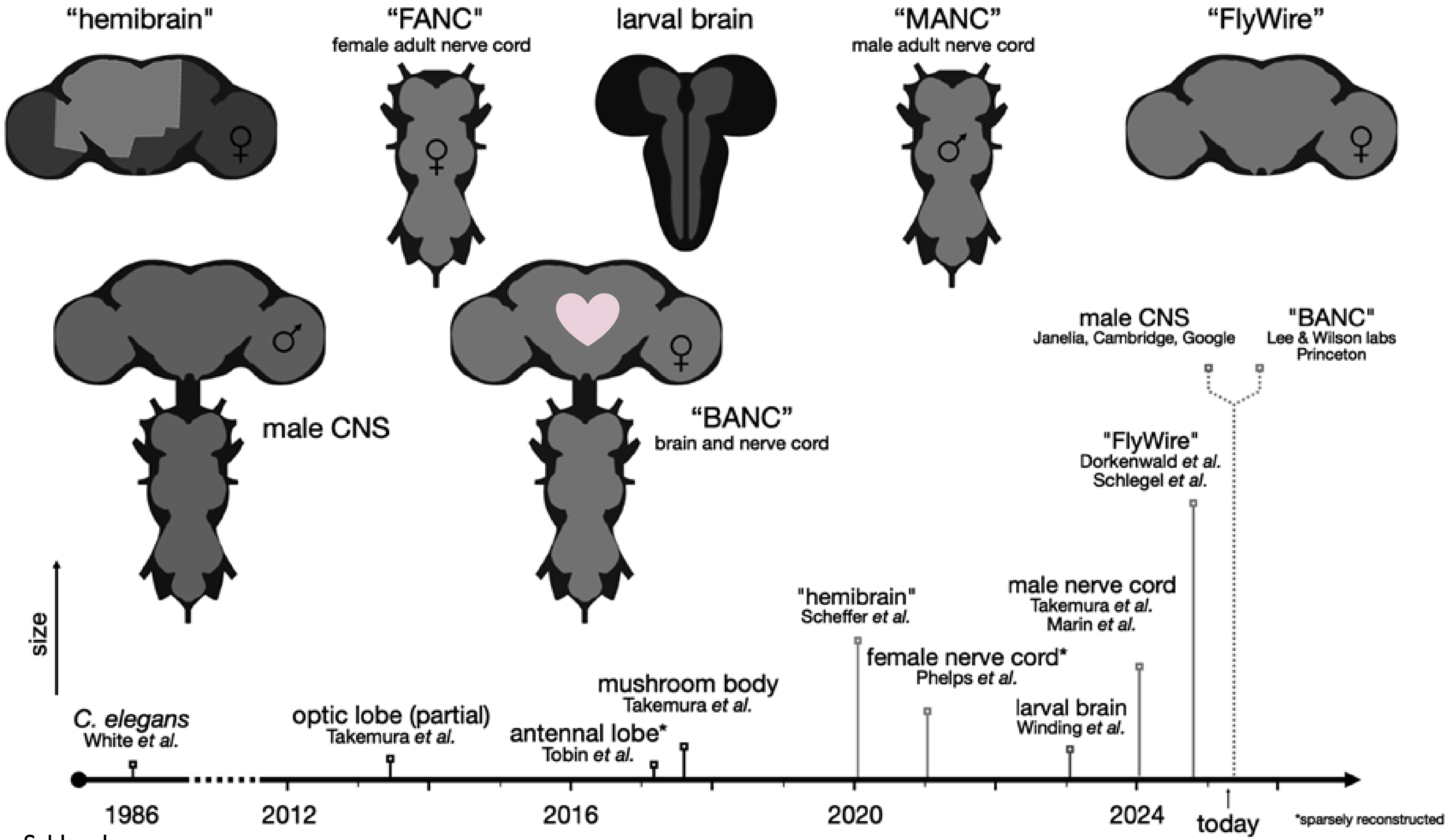
Connectomics Dataset

bioRxiv
THE PREPRINT SERVER FOR BIOLOGY

ng.banc.community/view

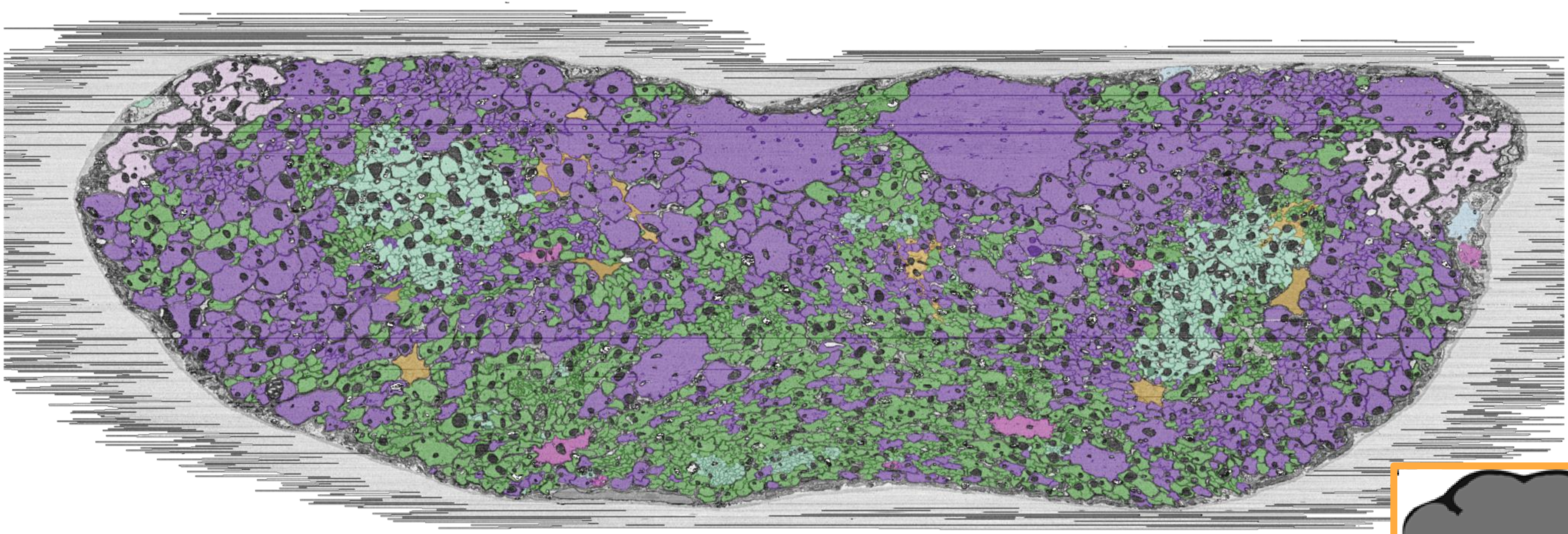
Connectome Explorer:

codex.flywire.ai/banc



Philipp Schlegel

transection of the neck connective



descending neurons
ascending neurons

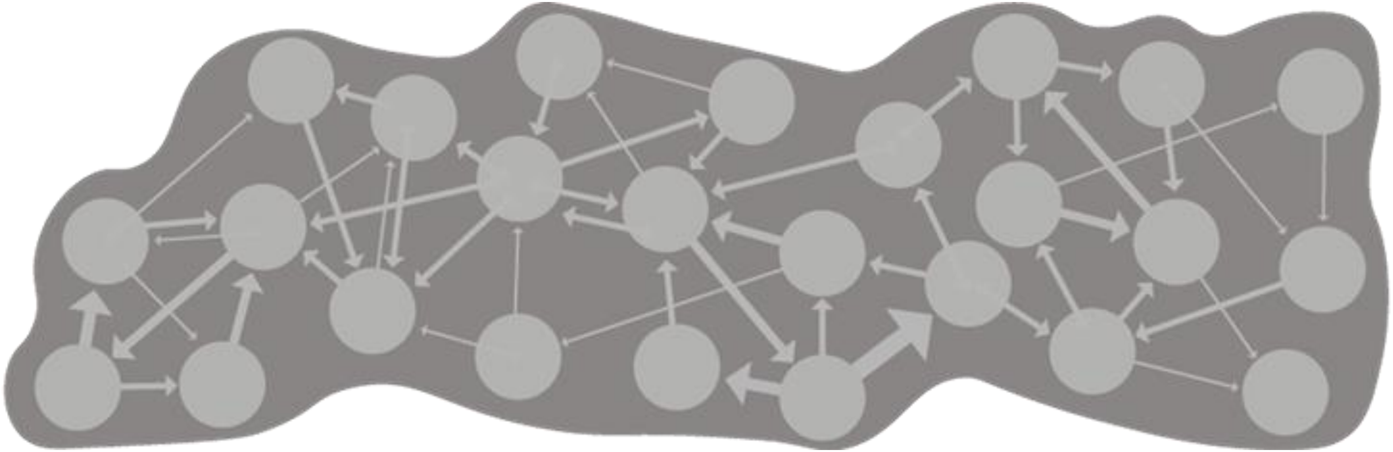
turning raw EM data into a *useful* connectome

a

all the neurons in the central nervous system reconstructed (~160k)

b

synapses detected across the whole volume (~20-100M)



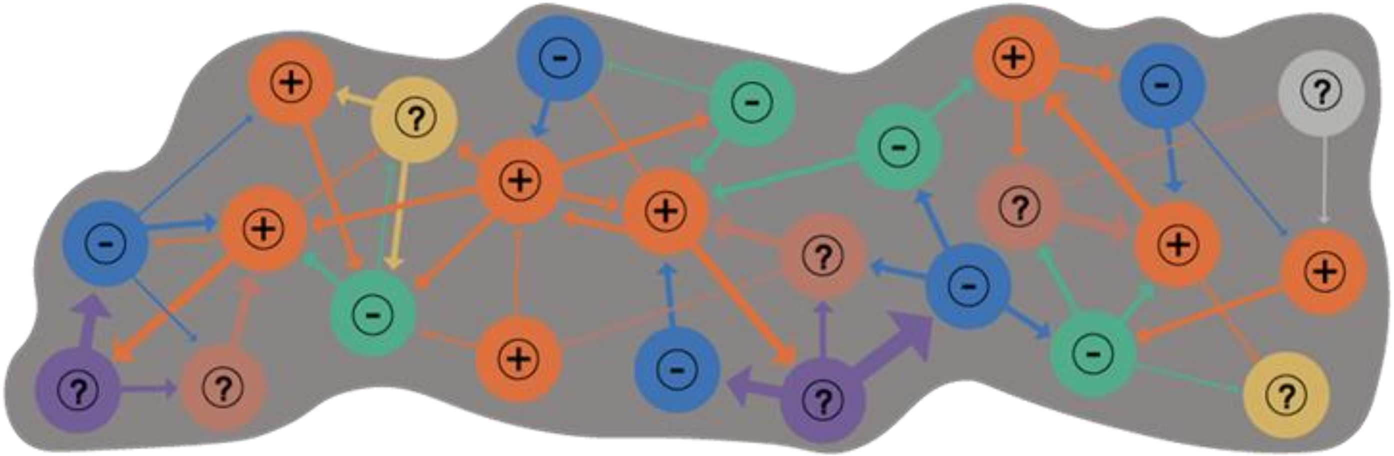
essential

turning raw EM data into a *useful* connectome

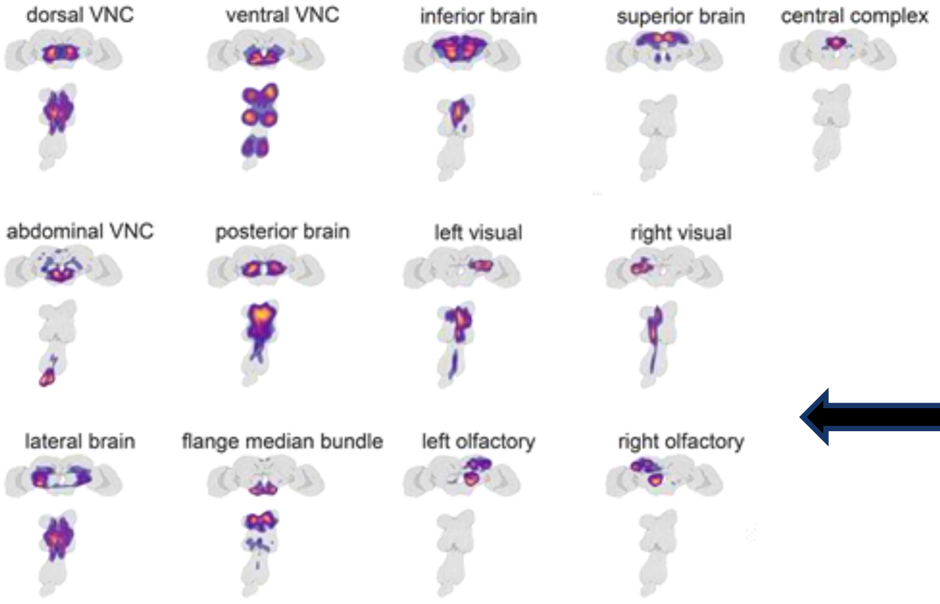
c **network signs**
i.e. neurotransmitter predictions

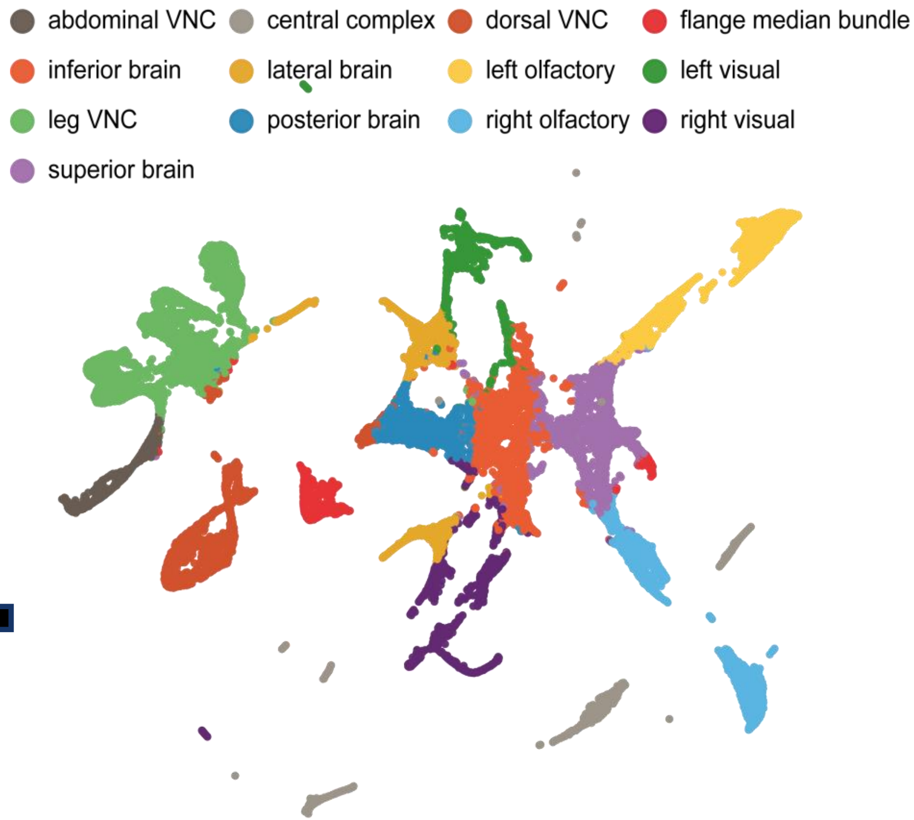
d **other detections**
e.g. neuropeptides

e **classification**
i.e. literature-consistent cell types

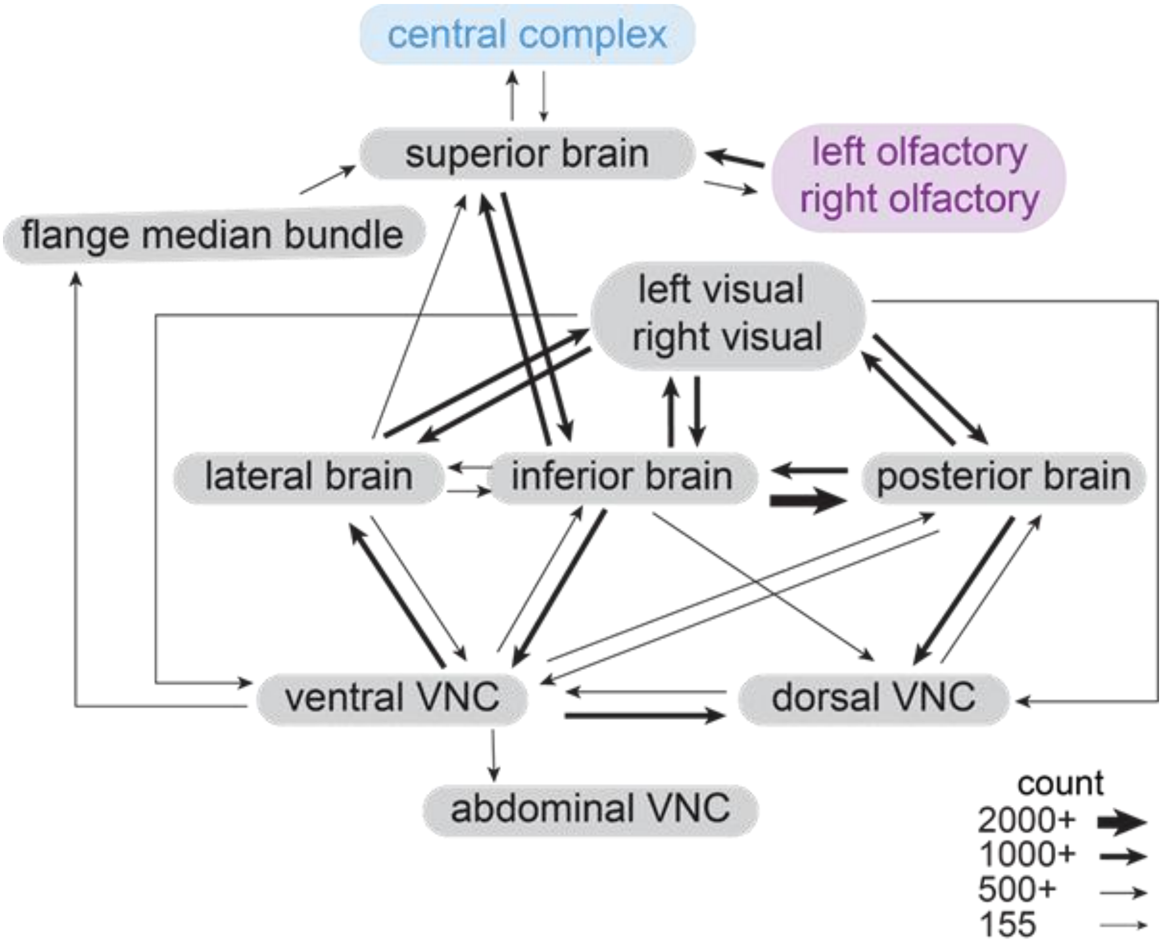


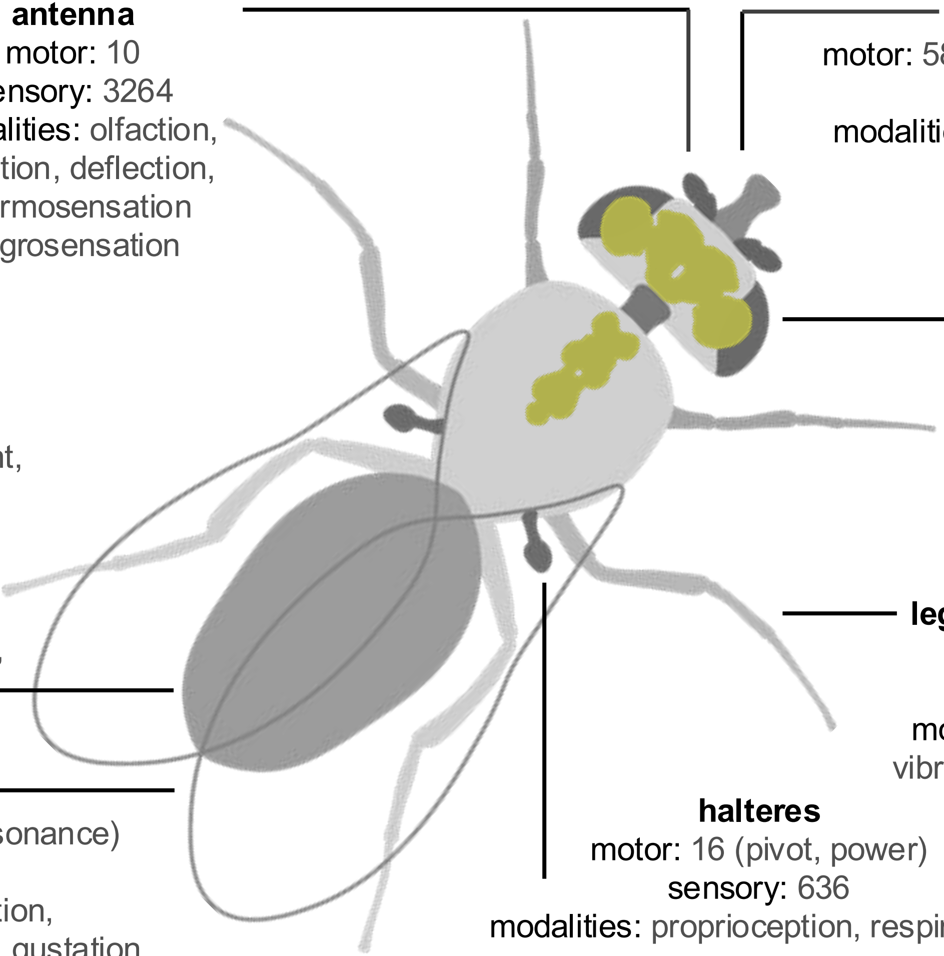
clustering neurons across the CNS





general overview of the CNS



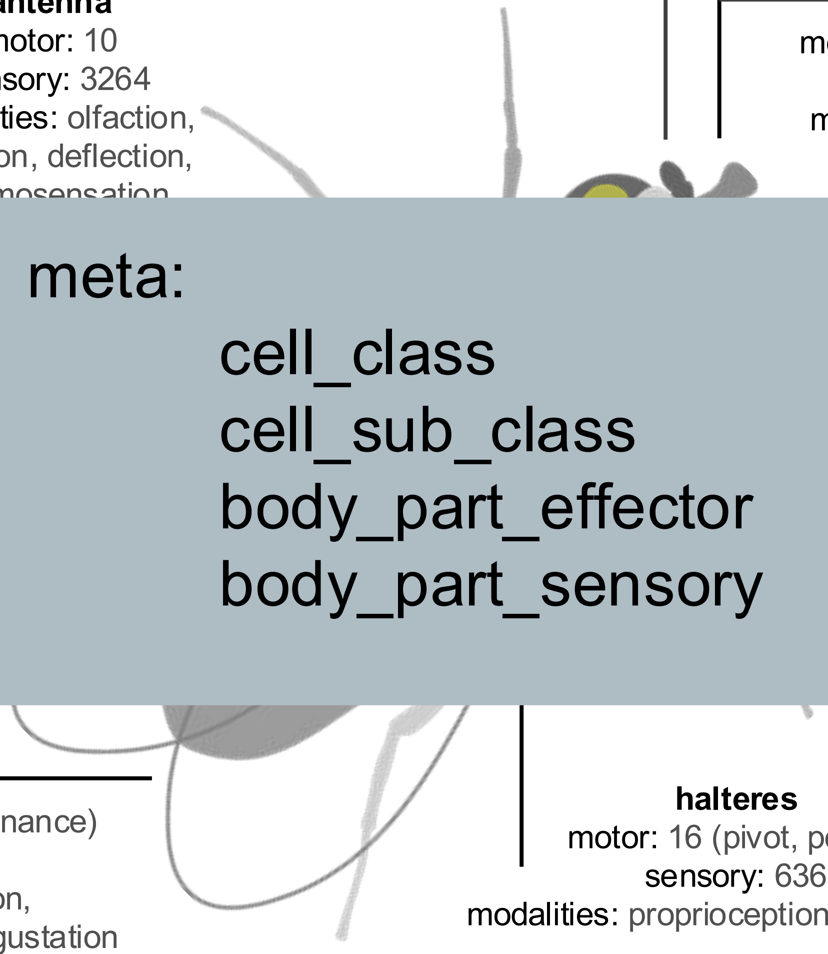


Anthony Azevedo, Helen Yang, Meet Zandawala, Stefanie Hampel, Andrew Seeds, Serene Dhawan, Sweta Agrawal, + more

other: retrocerebral complex, neurohemal organ, heart, crop
endocrine: ~90 (systemic hormones, internal o
motor: ~9
sensory: unl

antenna
motor: 10
sensory: 3264
modalities: olfaction, vibration, deflection, thermosensation

proboscis
motor: 58 (moving, ingesting)
sensory: 58
modalities: gustation, tactile



meta:

cell_class
cell_sub_class
body_part_effector
body_part_sensory

eye
tor: 4
ry: 1,110
hromatic vision,
c vision, UV,
arised

motor: 195
defecati
er
se
modalitie
vibration,
g

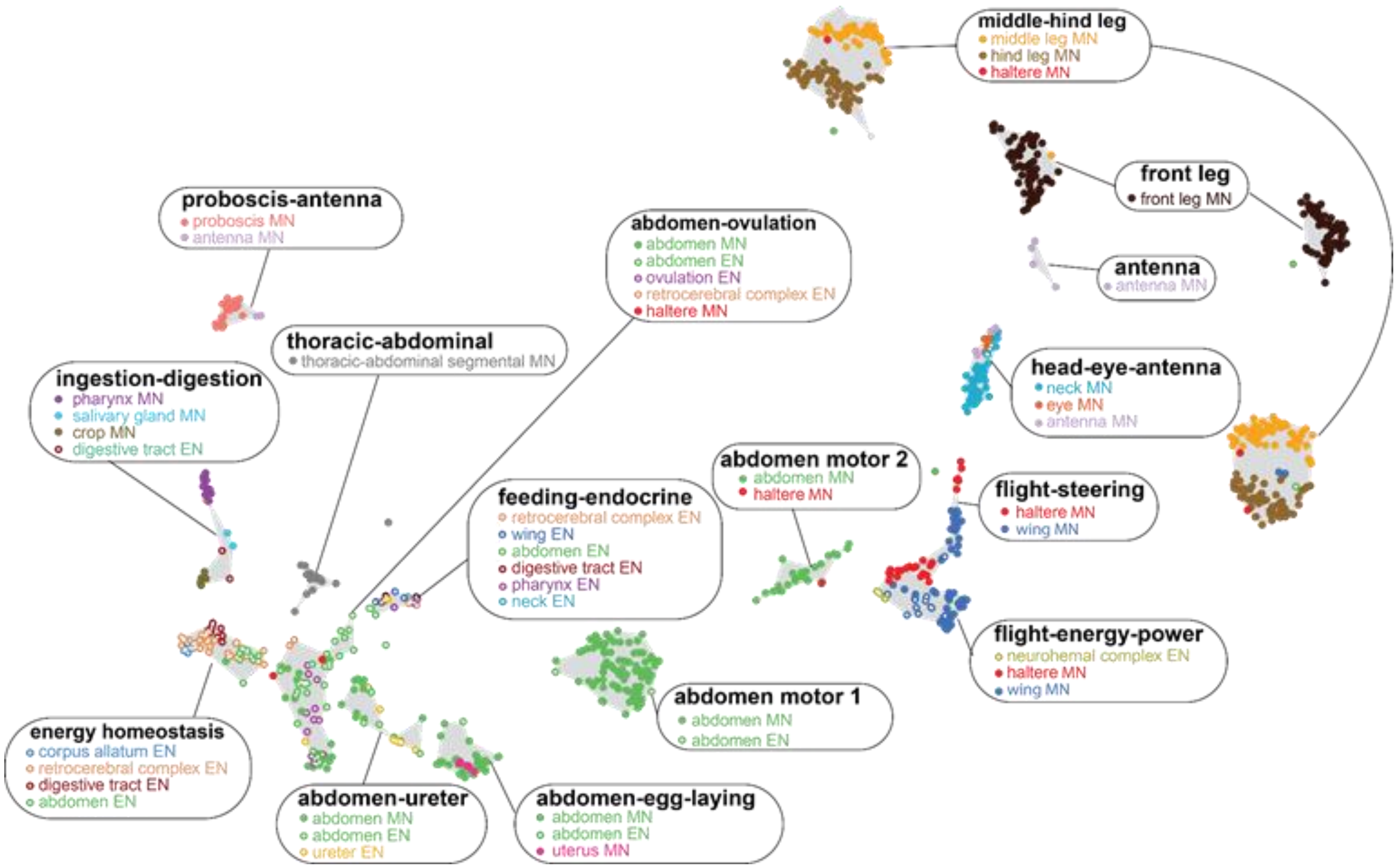
ddle, hind)
396
y: 3678
modalities: proprioception, vibration, tactile, nociception, gustation

wings
motor: 82 (pivot, power, resonance)
sensory: 989
modalities: proprioception, vibration, tactile, nociception, gustation

halteres
motor: 16 (pivot, power)
sensory: 636
modalities: proprioception, respiration,

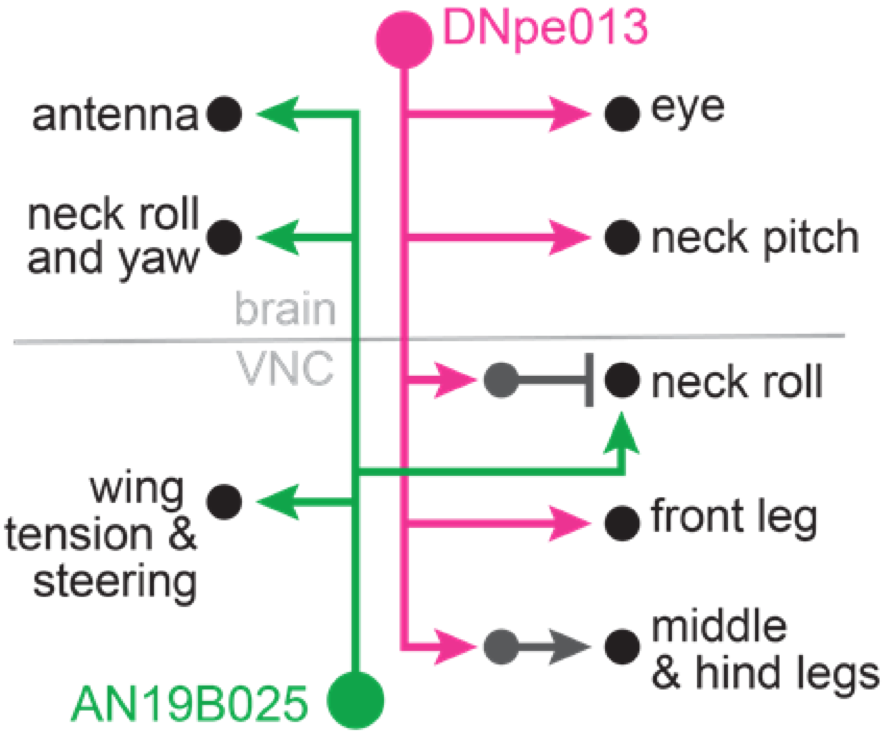
Anthony Azevedo, Helen Yang, Meet Zandawala, Stefanie Hampel, Andrew Seeds, Serene Dhawan, Sweta Agrawal, + more

common upstream partners define sets of effectors

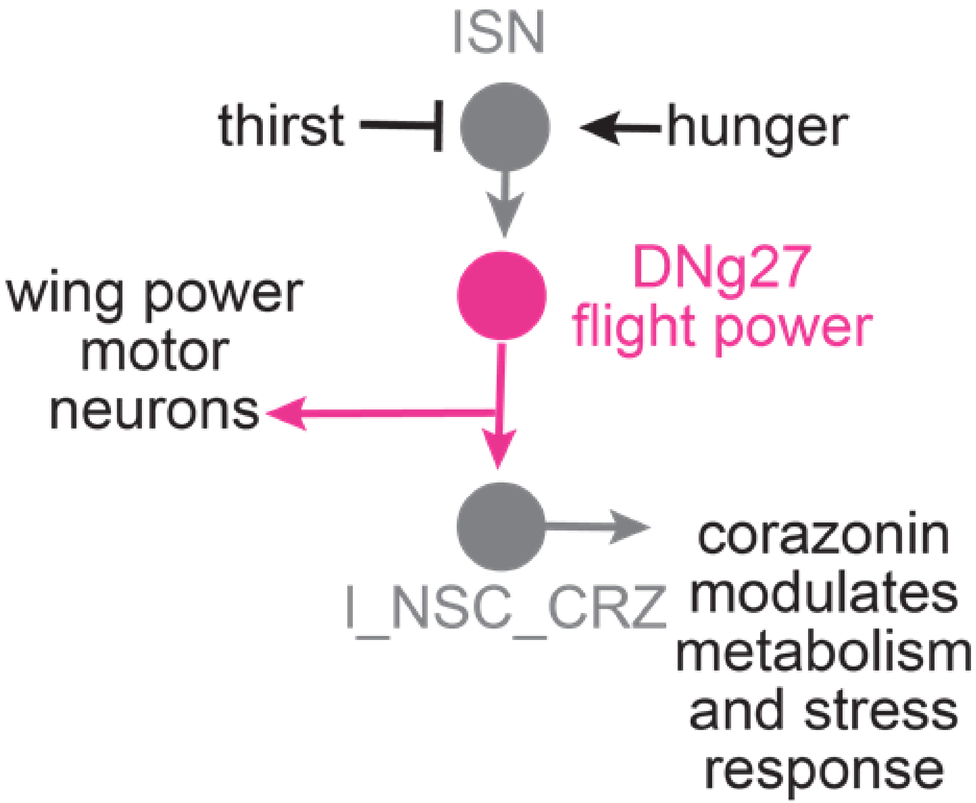


ascending and descending neurons link effectors across the body

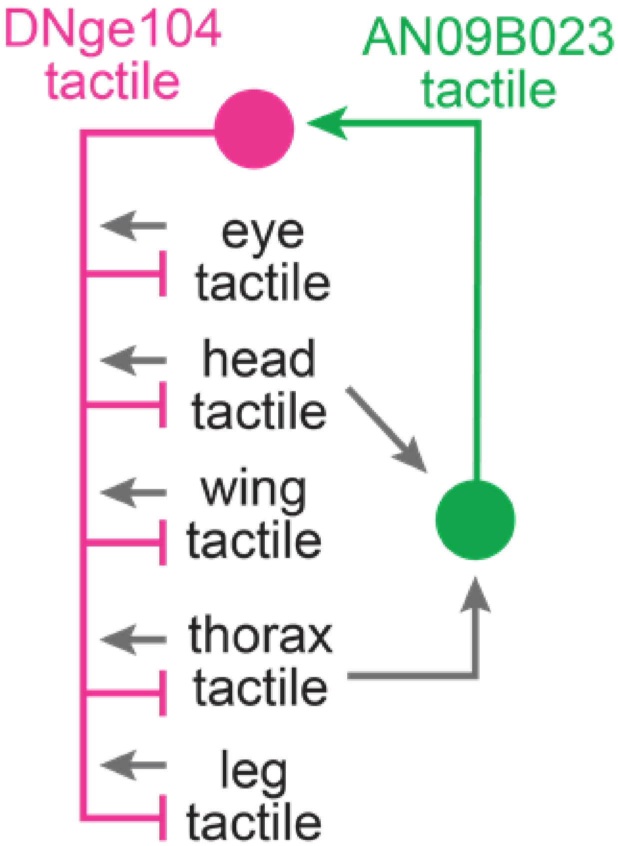
circuit: directing the body



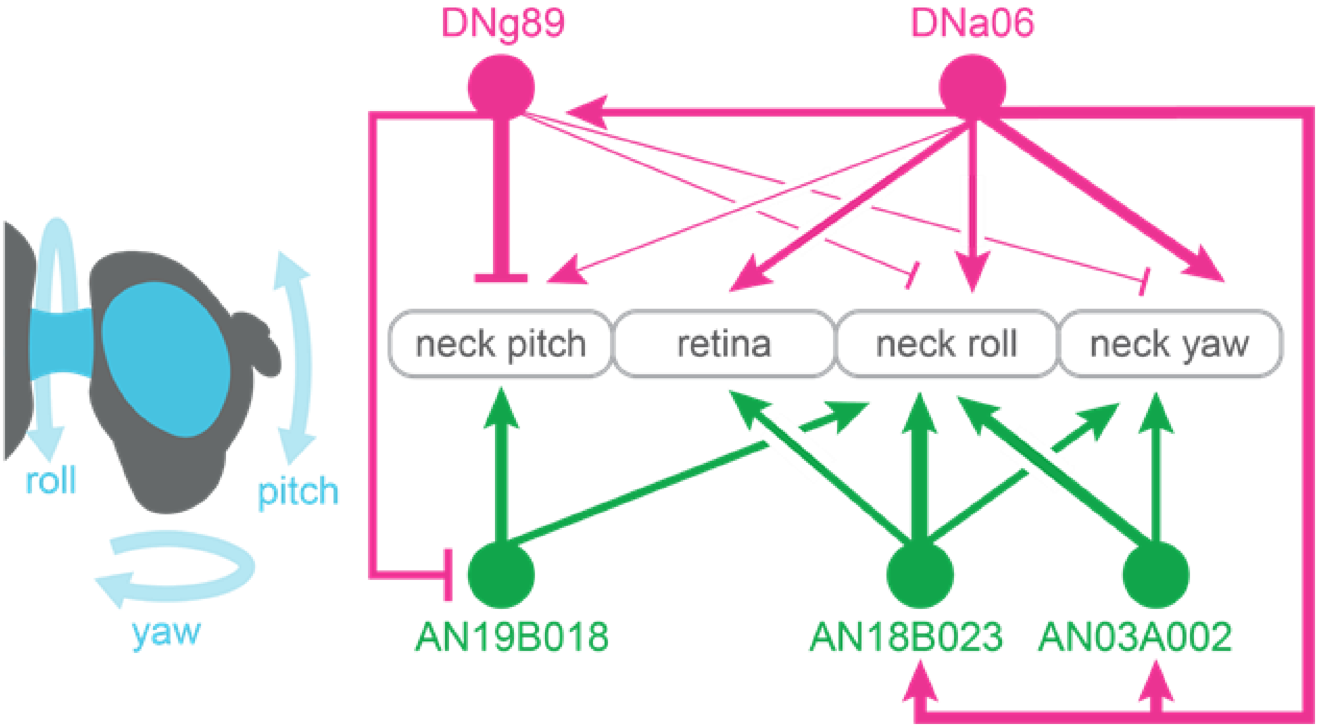
circuit: co-regulation of flight power and endocrine stress response



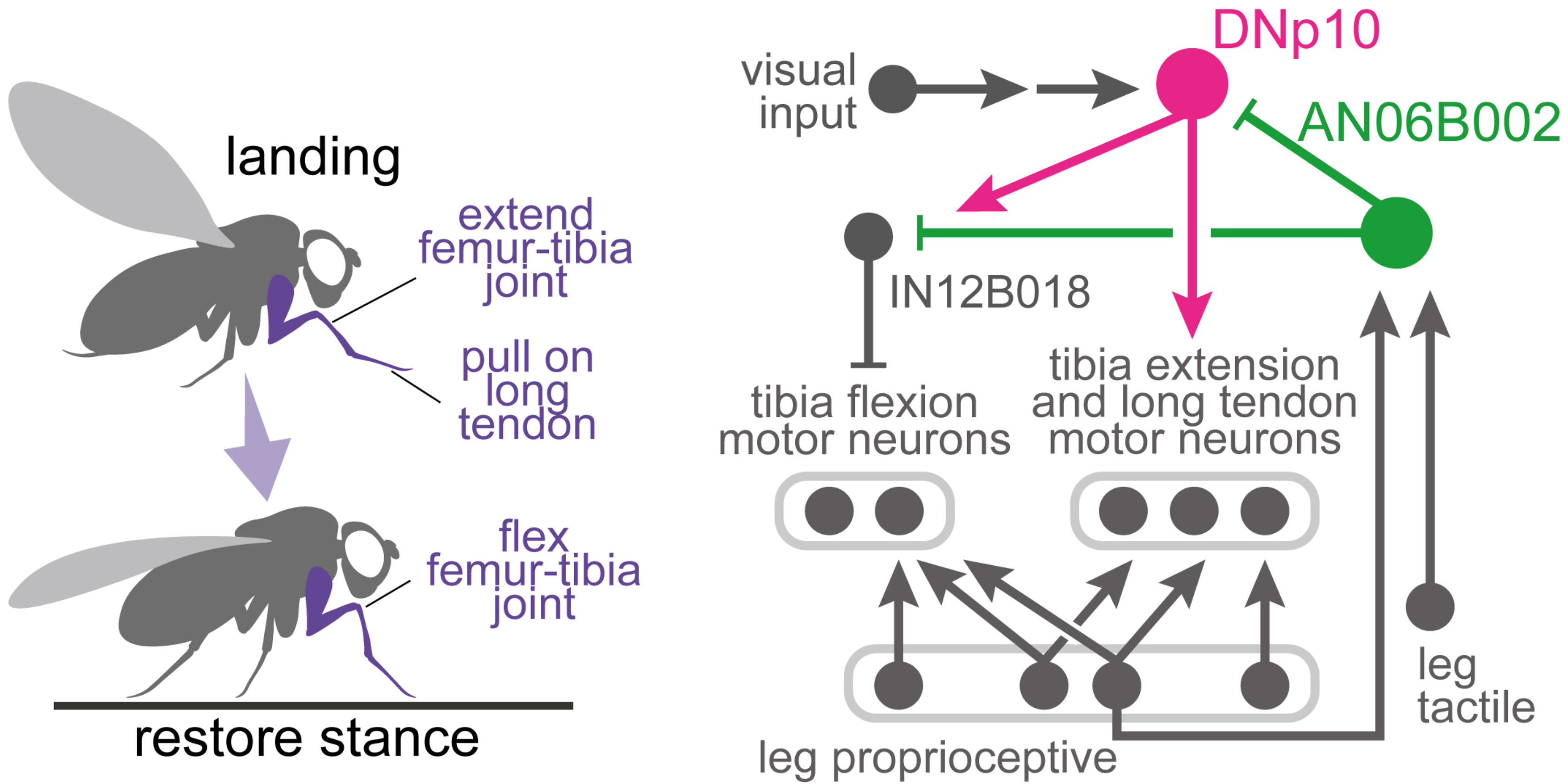
circuit: sensory contrast enhancement



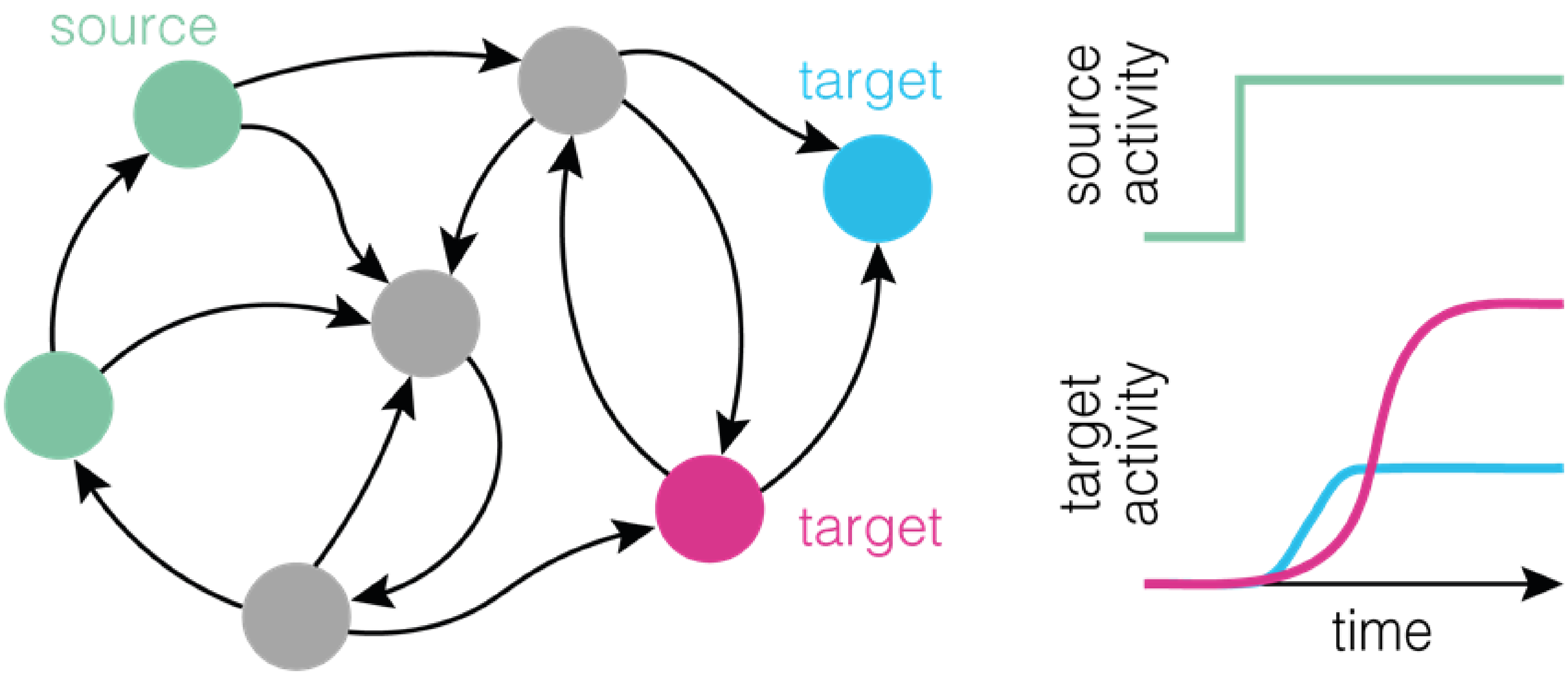
circuit: pitching, rolling and yawing the neck



circuit: landing



The "influence" score



https://github.com/DrugowitschLab/ConnectomeInfluenceCalculator

https://github.com/natverse/influencer

linear dynamical model

$$\tau \frac{d\boldsymbol{r}(t)}{dt} = -\boldsymbol{r}(t) + \boldsymbol{W}\boldsymbol{r}(t) + \boldsymbol{s}$$

time constant

neural activity
(160,000 x 1)

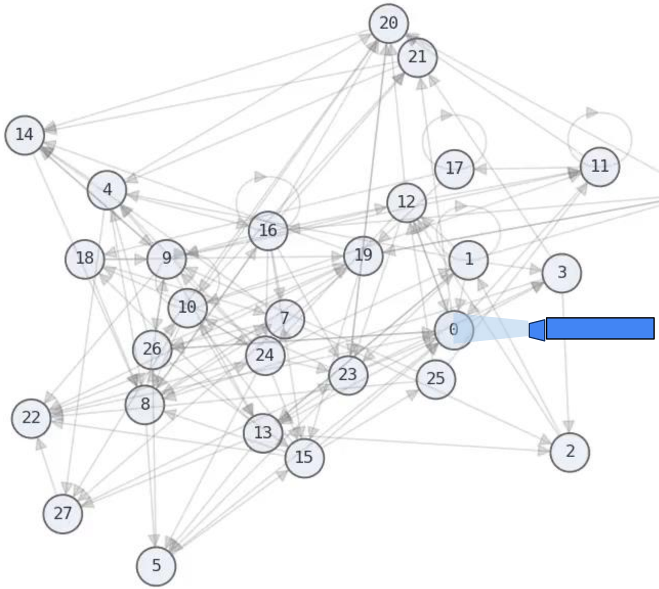
connectivity matrix
(160,000 x 160,000)

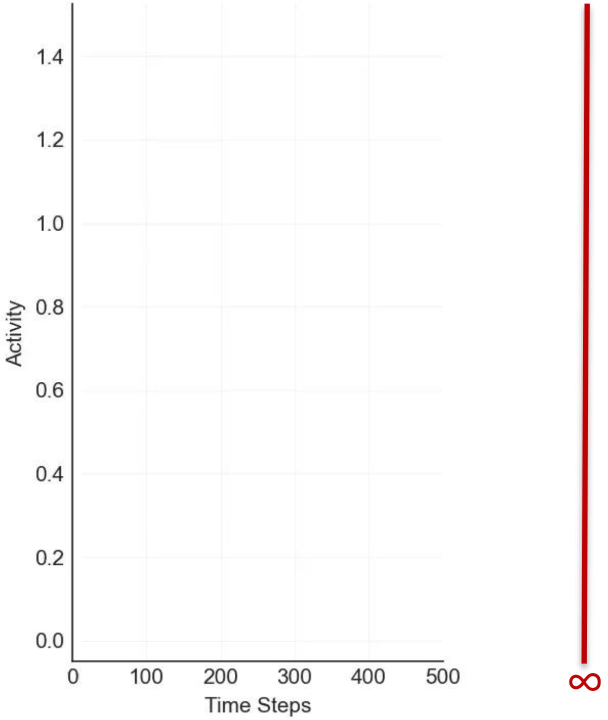
external input (Seed)
(160,000 x 1)

linear dynamical model

$$0 \quad = -\boldsymbol{r}(t) + \boldsymbol{W}\boldsymbol{r}(t) + \boldsymbol{s}$$

Steady-state





linear dynamical model

$$\tau \frac{d\boldsymbol{r}(t)}{dt} = -\boldsymbol{r}(t) + \boldsymbol{W}\boldsymbol{r}(t) + \boldsymbol{s}$$

pros:

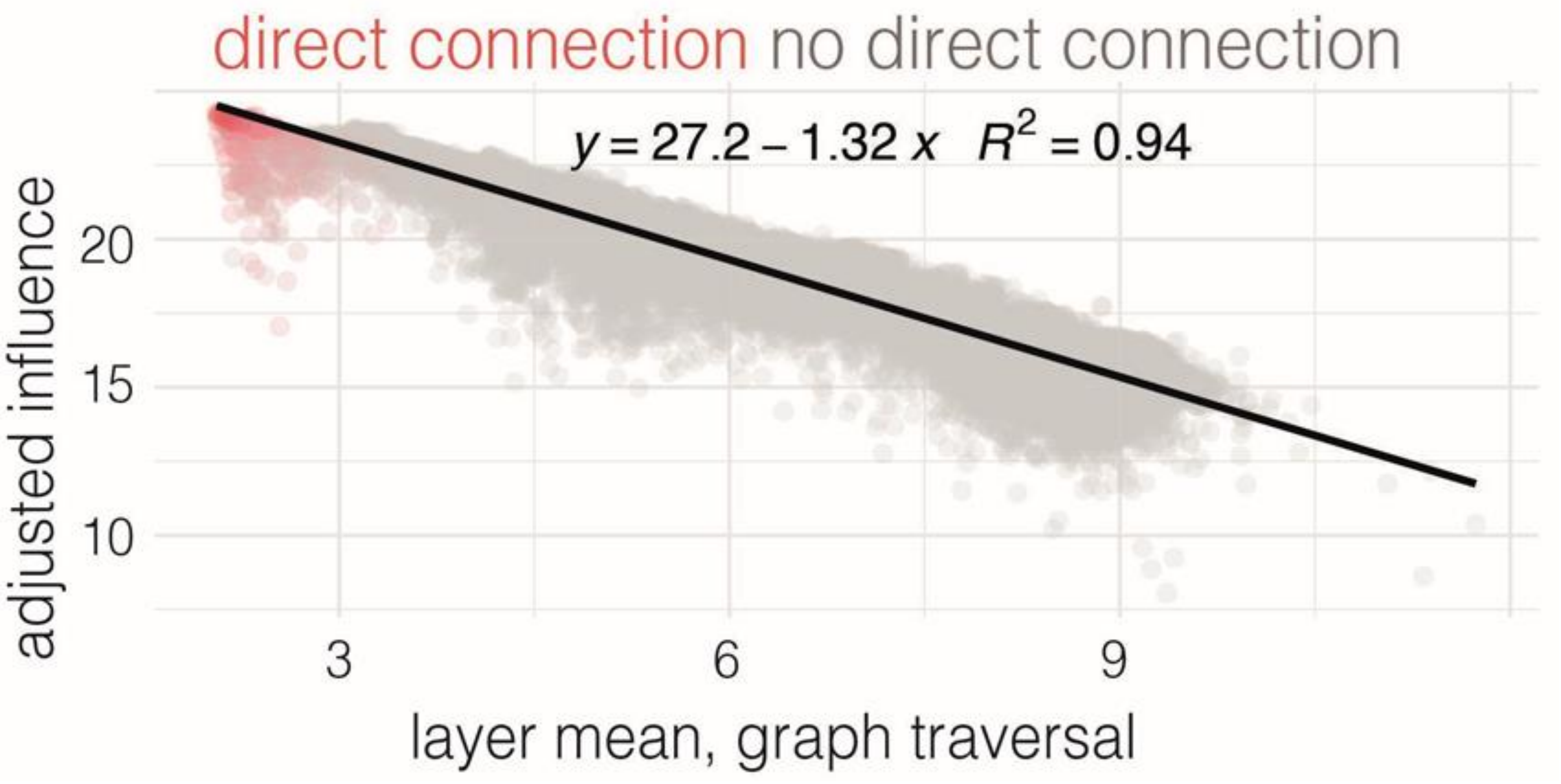
- Efficient and fast computation (~25 billion influence scores computed in less than 20 hours)
- Covers all possible pathways
- Additive method

cons:

- Simplistic model
- All synapse types are treated as excitatory

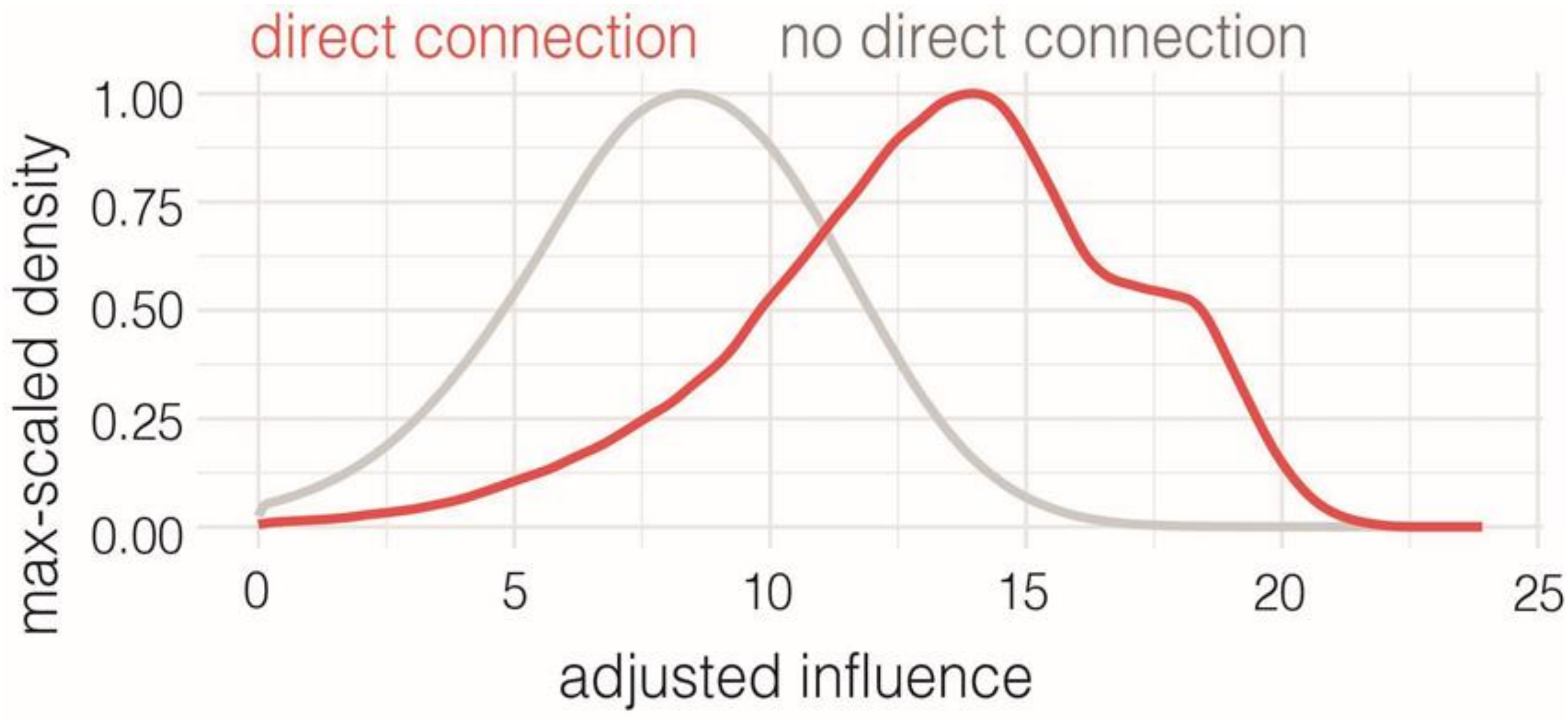
this is not a biophysical model

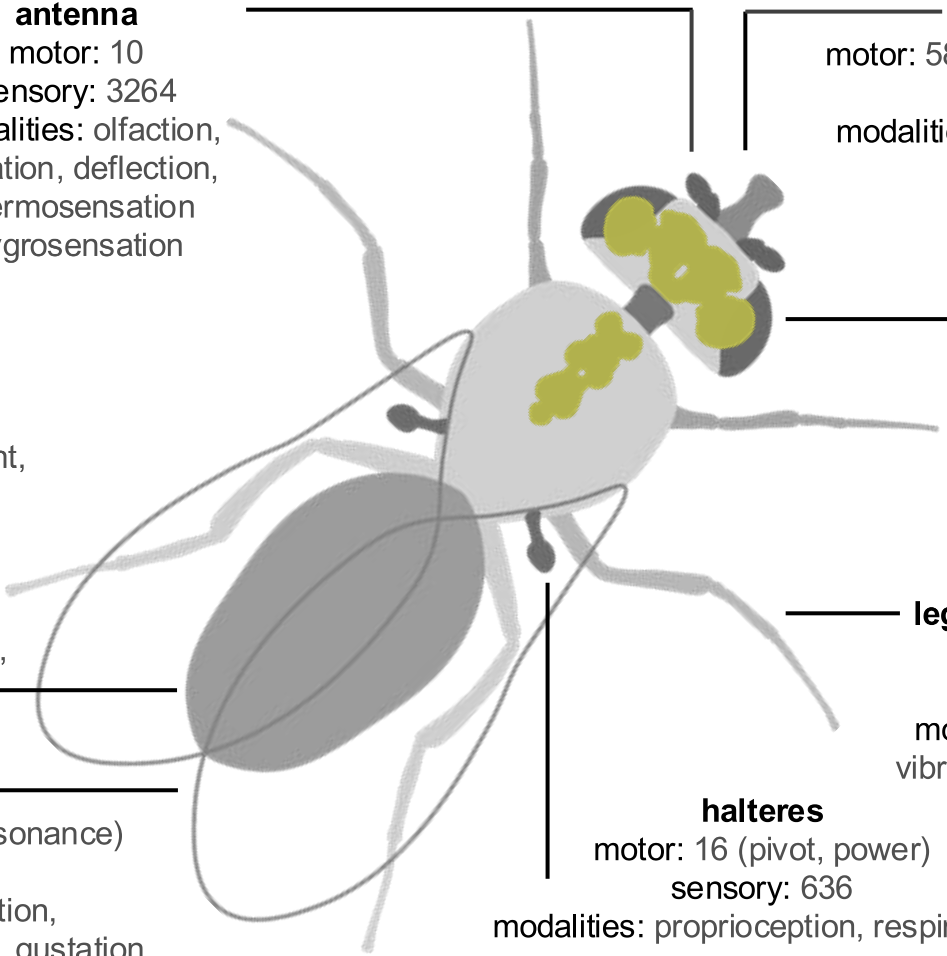
methods for interpreting the connectome



Schlegel and Bates et al., 2021

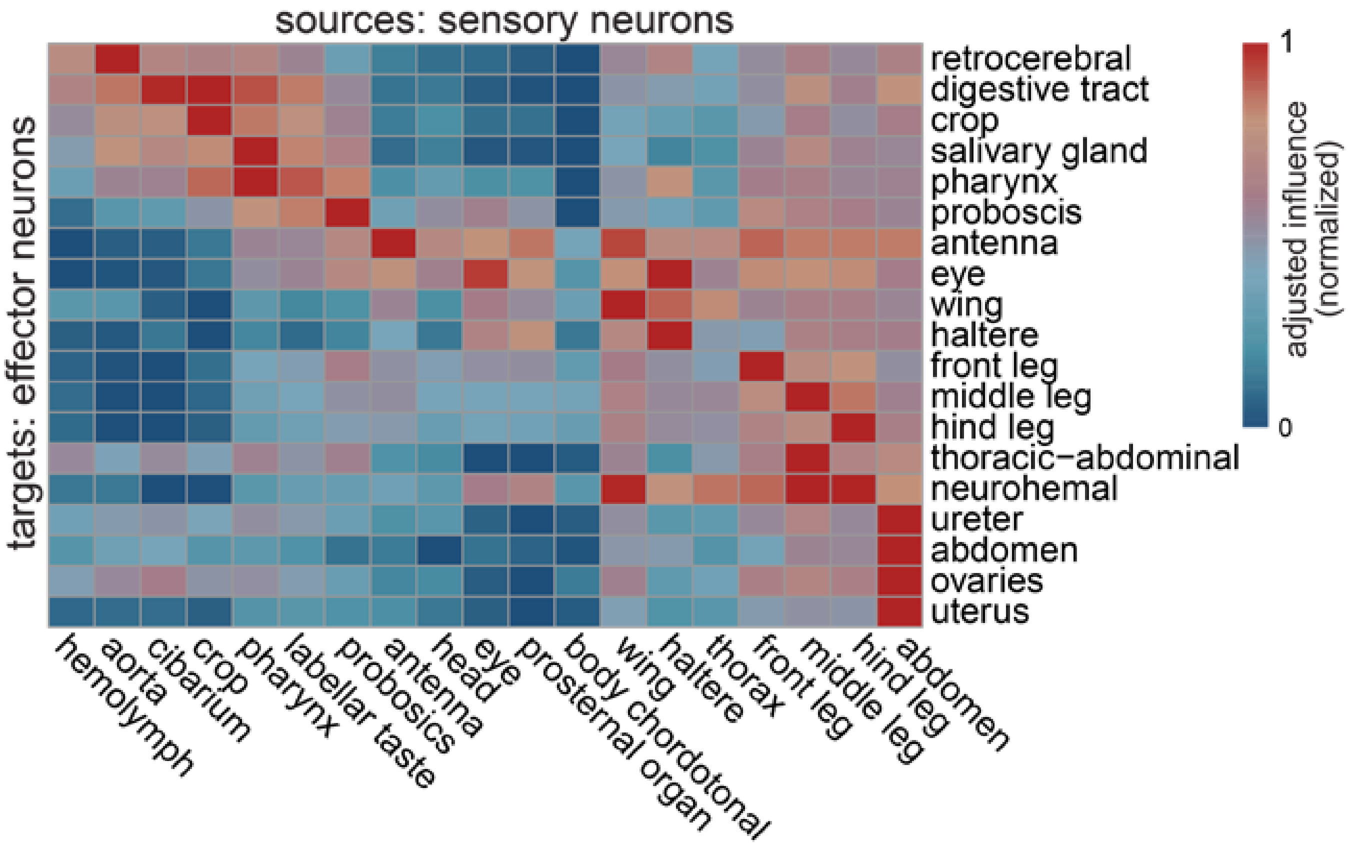
methods for interpreting the connectome





Anthony Azevedo, Helen Yang, Meet Zandawala, Stefanie Hampel, Andrew Seeds, Serene Dhawan, Sweta Agrawal, + more

sensors mainly influence effectors of the same body part
but there is also long-range influence



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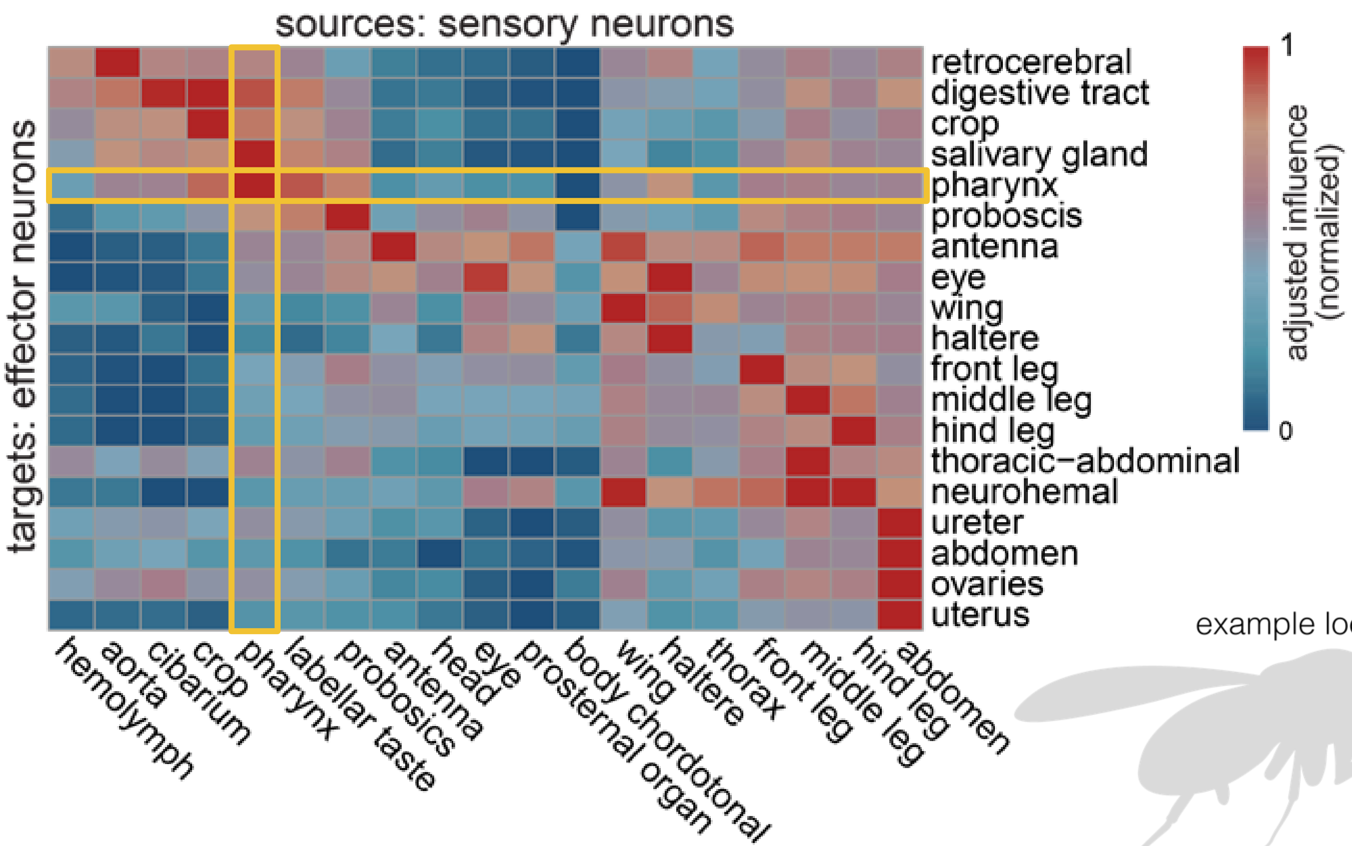

sensors mainly influence effectors of the same body part
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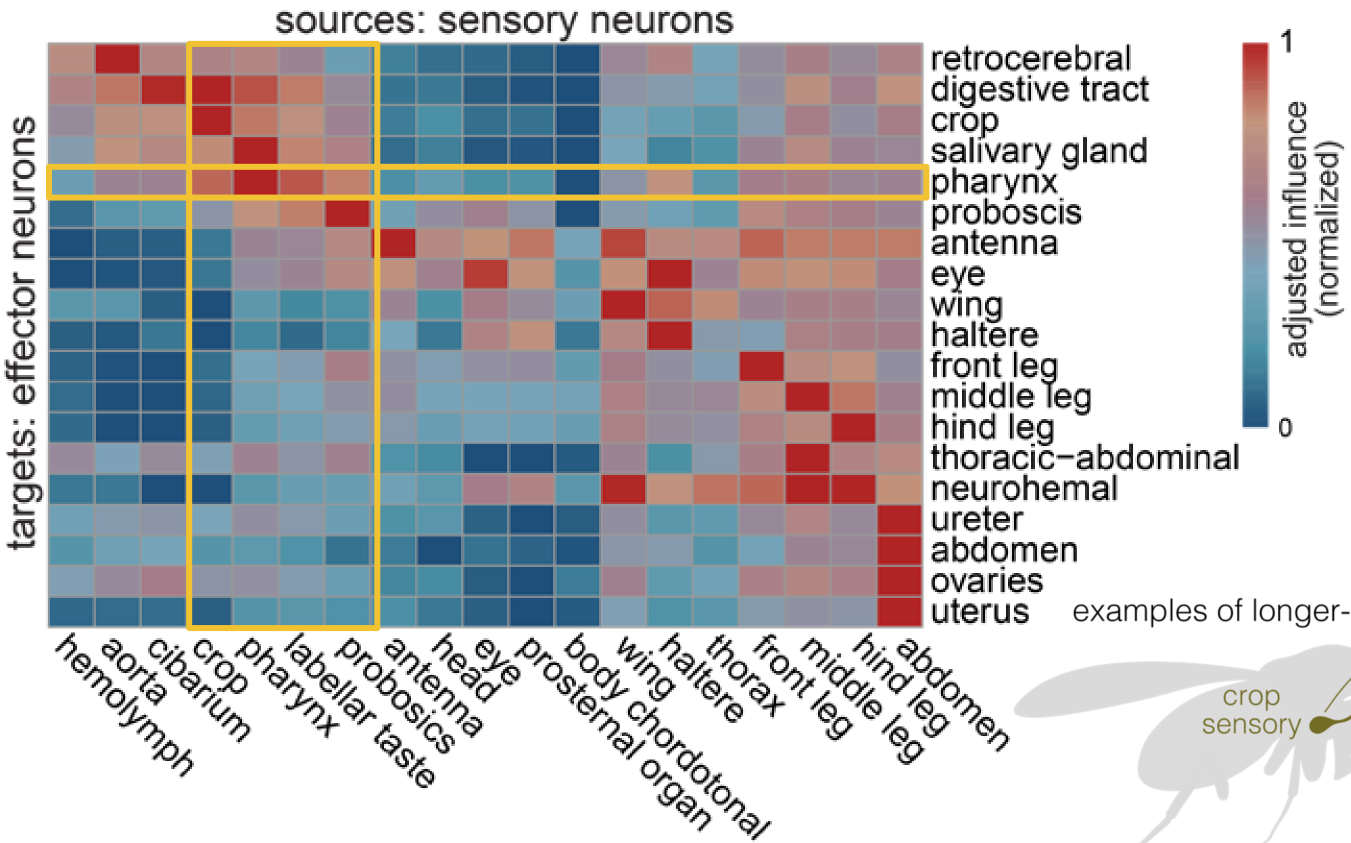
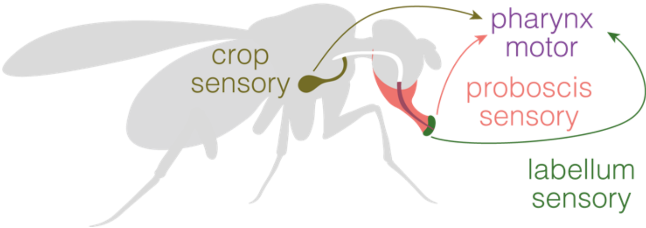
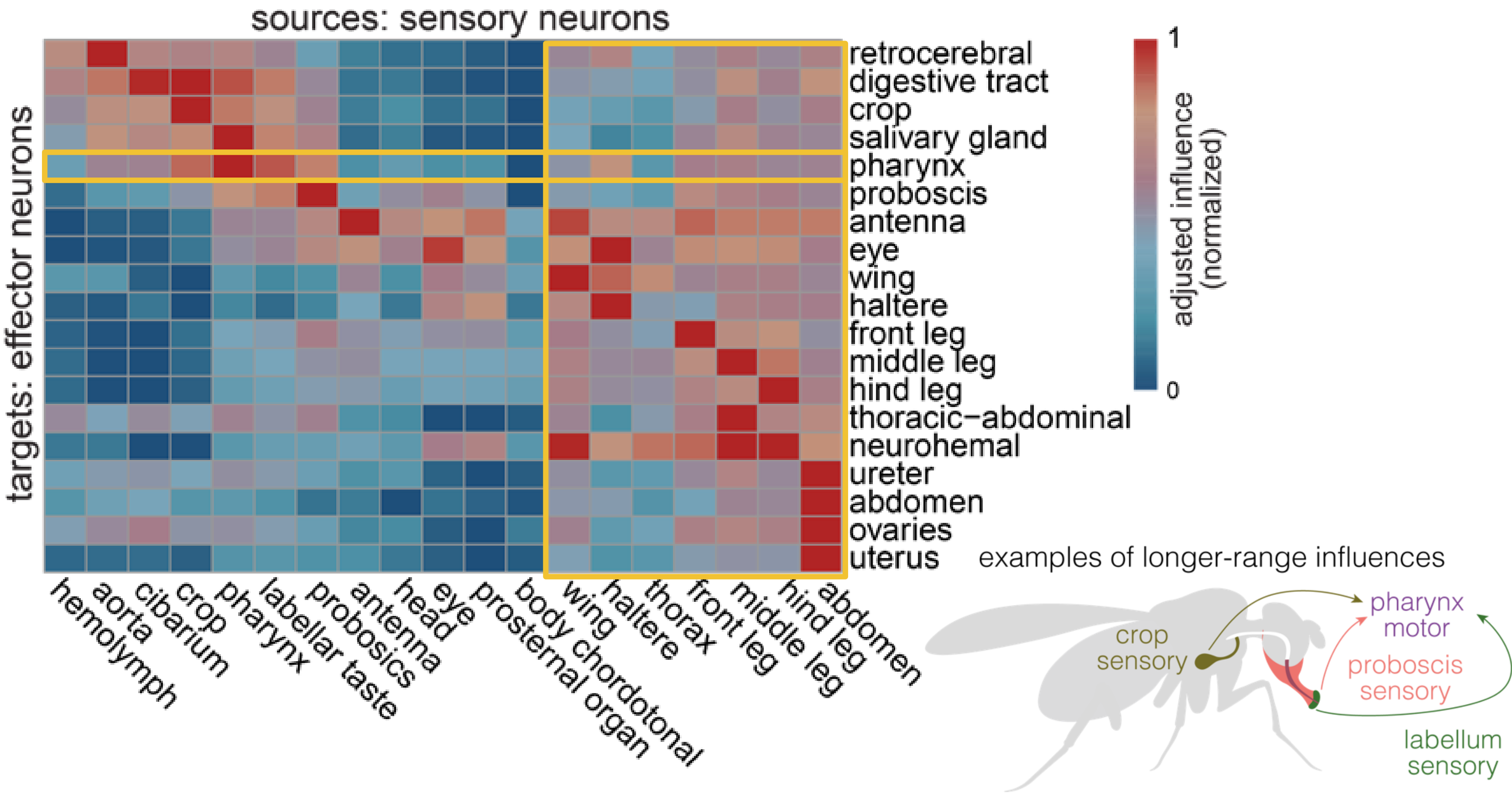


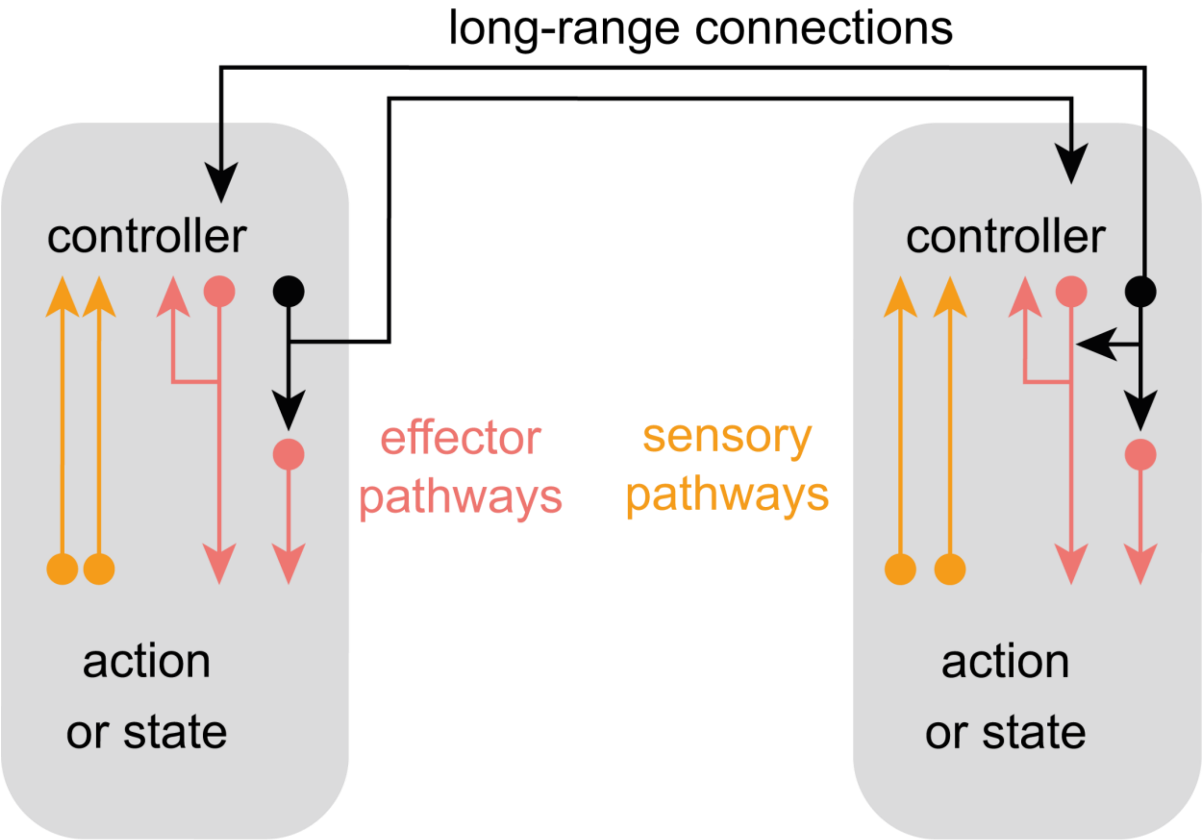
examples of longer-range influences



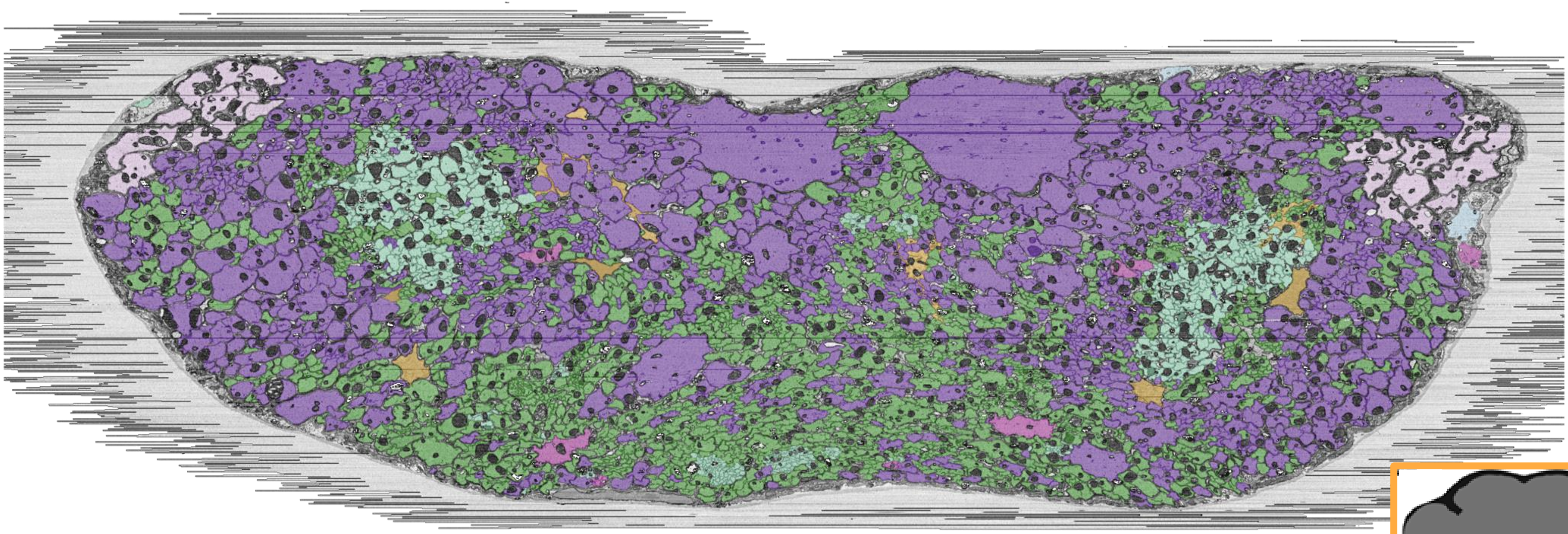
sensors mainly influence effectors of the same body part
but there is also long-range influence



effector neurons have local circuits that are chained at higher levels



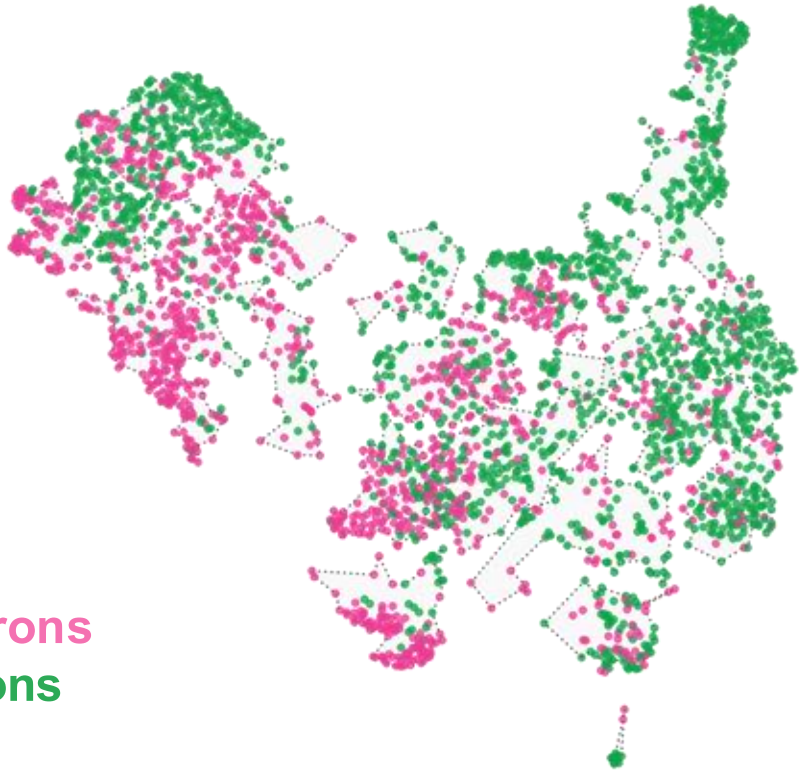
long-ranged connections transfer sensorimotor information across the CNS



descending neurons

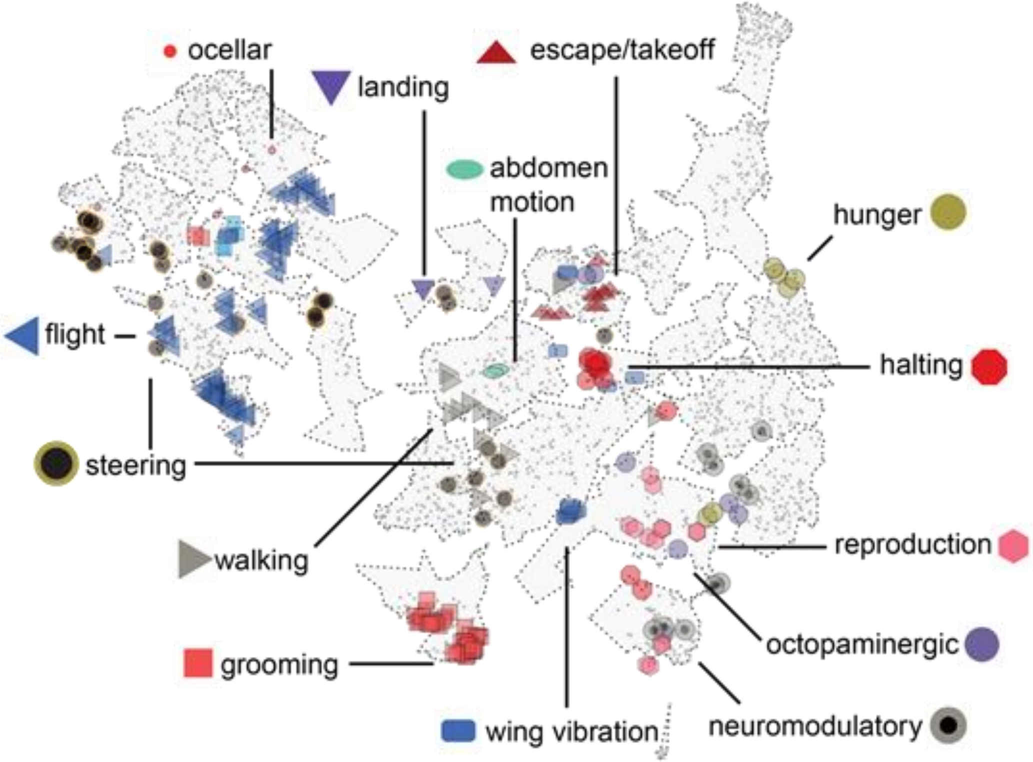
ascending neurons

ascending and descending neurons link sets of sensors to sets of effectors

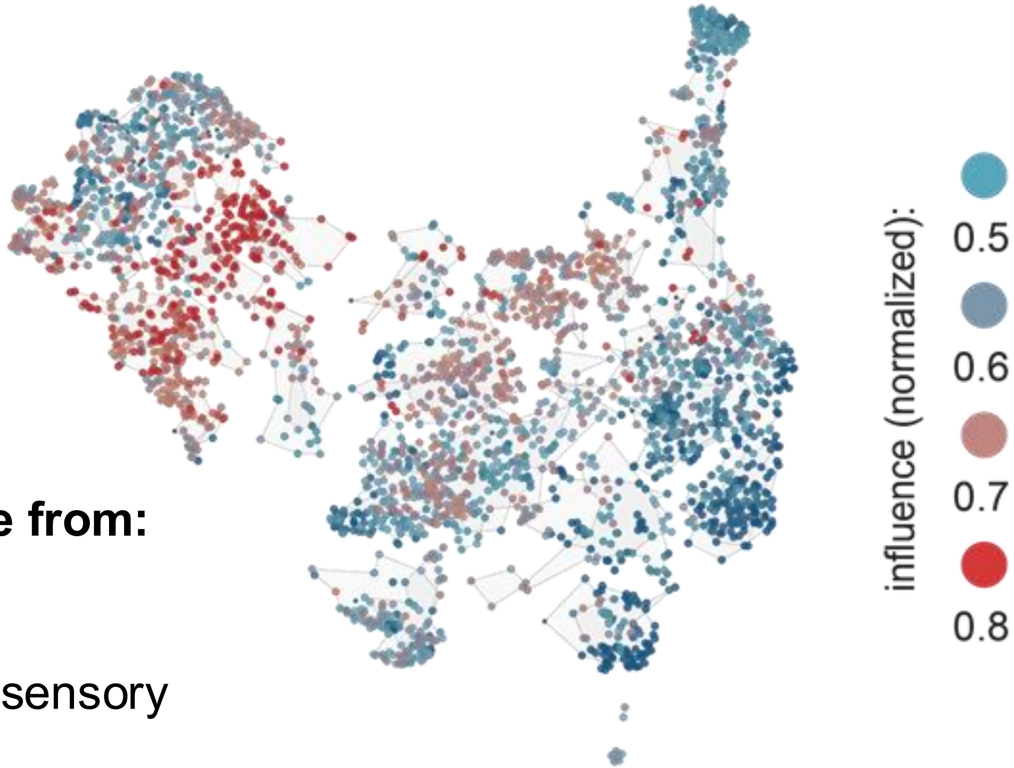


descending neurons
ascending neurons

previously characterized ascending and descending neurons cluster together

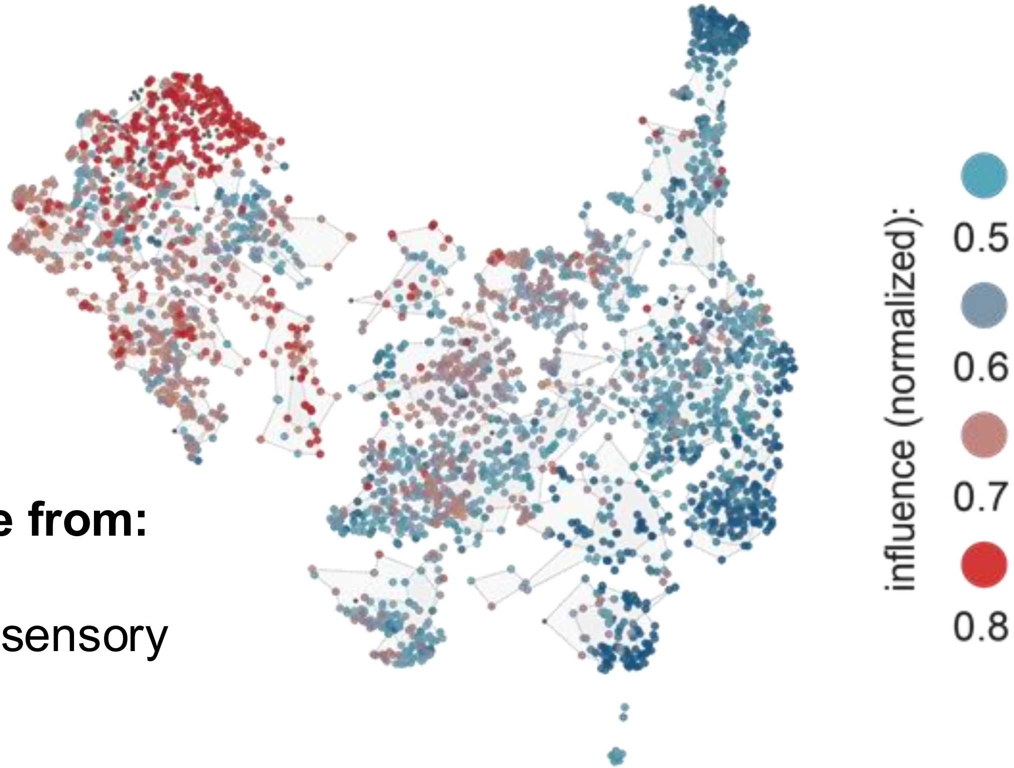


ascending and descending neurons link sets of sensors to sets of effectors



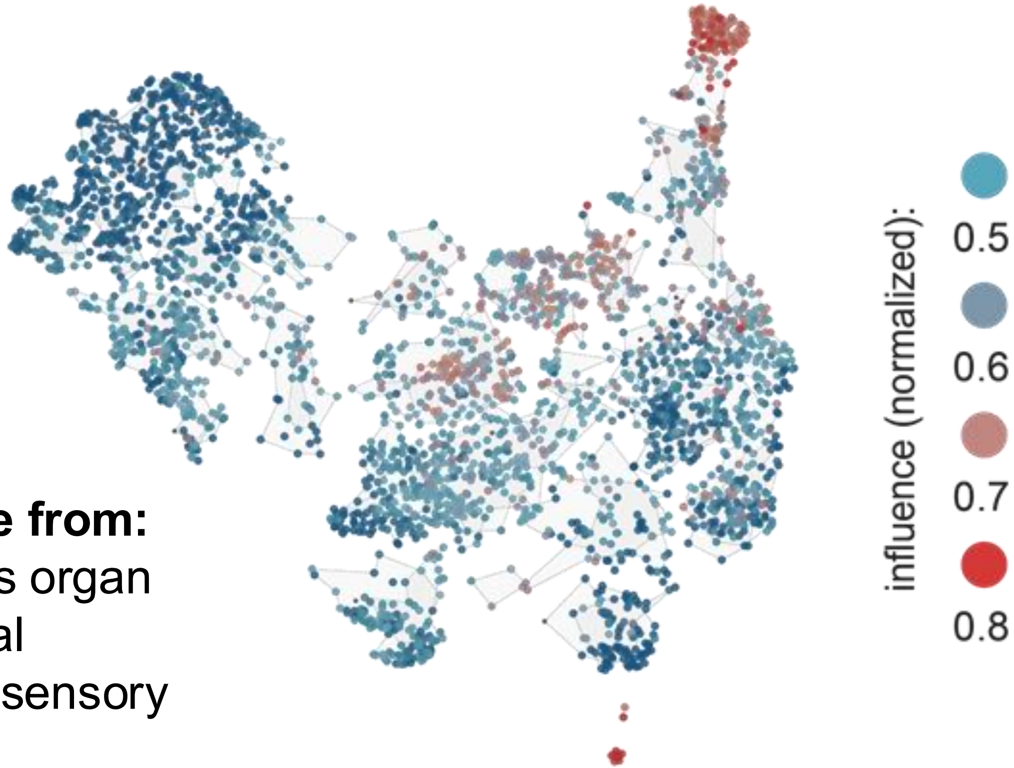
influence from:
JO-E
antenna
mechanosensory

ascending and descending neurons link sets of sensors to sets of effectors



influence from:
haltere
mechanosensory

ascending and descending neurons link sets of sensors to sets of effectors

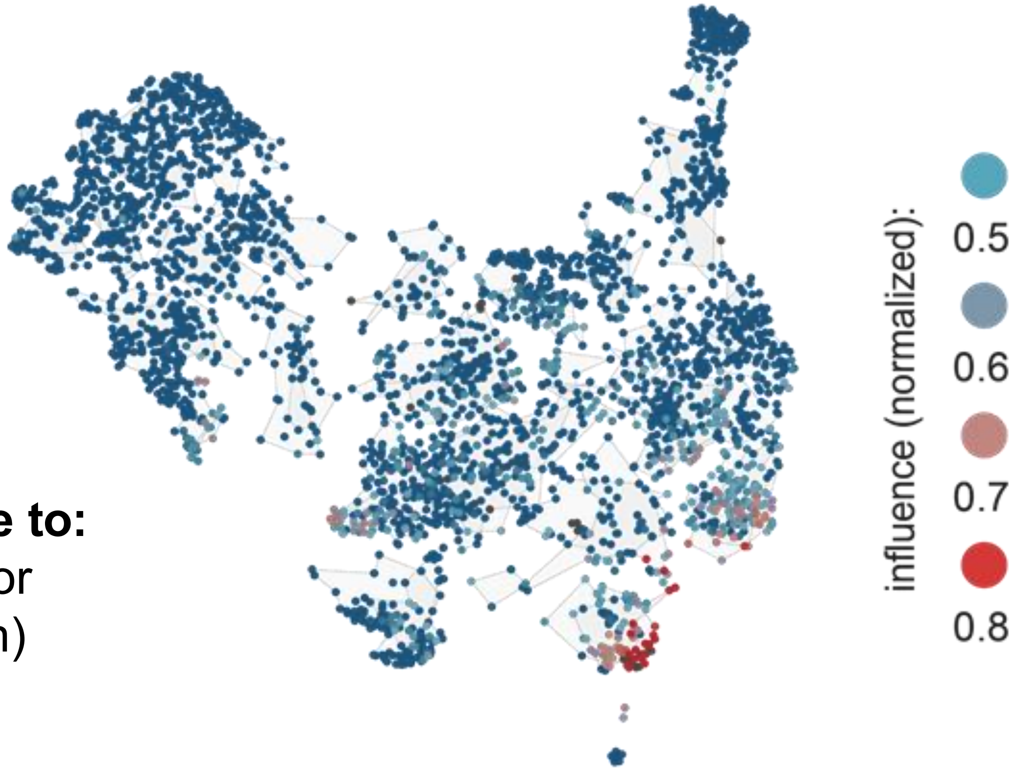


influence from:
Wheeler’s organ
abdominal
mechanosensory

ascending and descending neurons link sets of sensors to sets of effectors

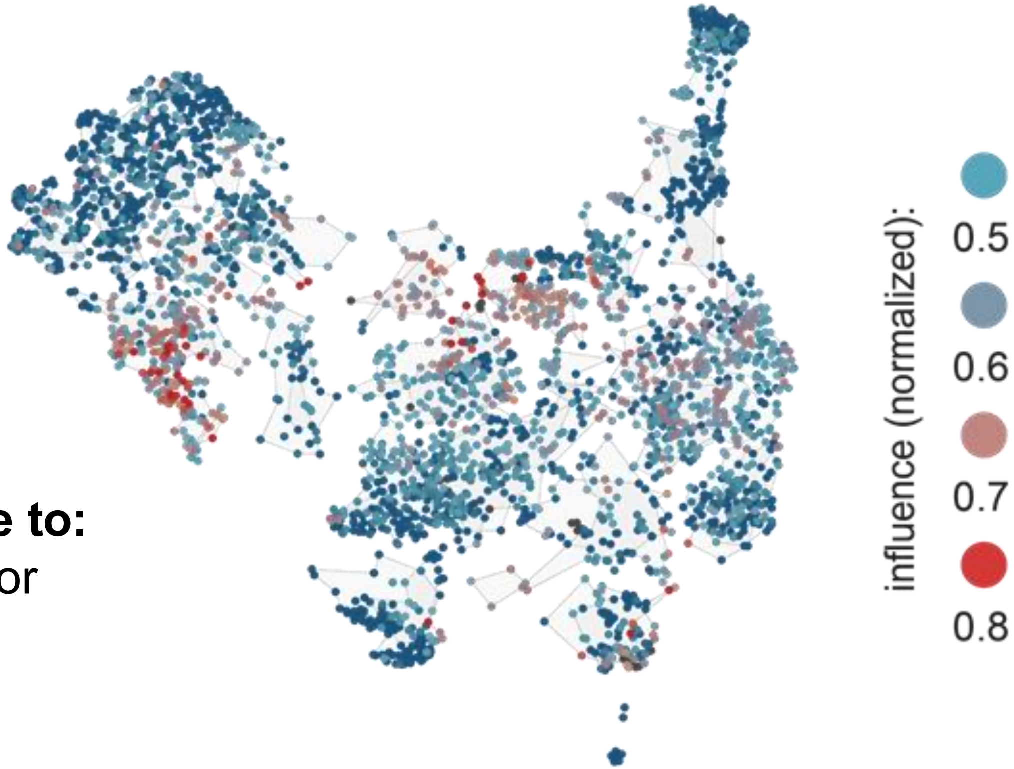
influence to:
front leg motor

ascending and descending neurons link sets of sensors to sets of effectors



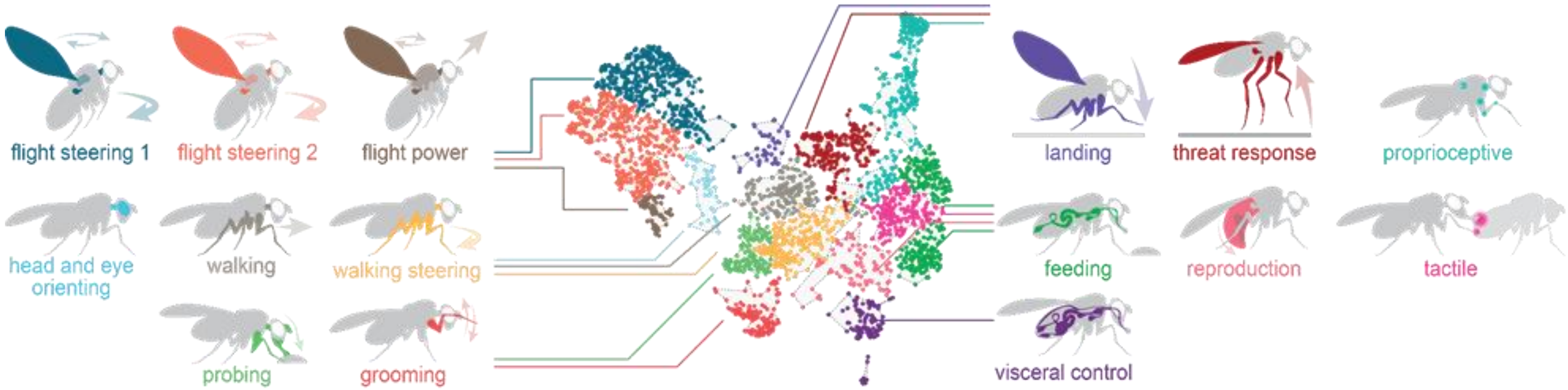
influence to:
crop motor
(ingestion)

ascending and descending neurons link sets of sensors to sets of effectors

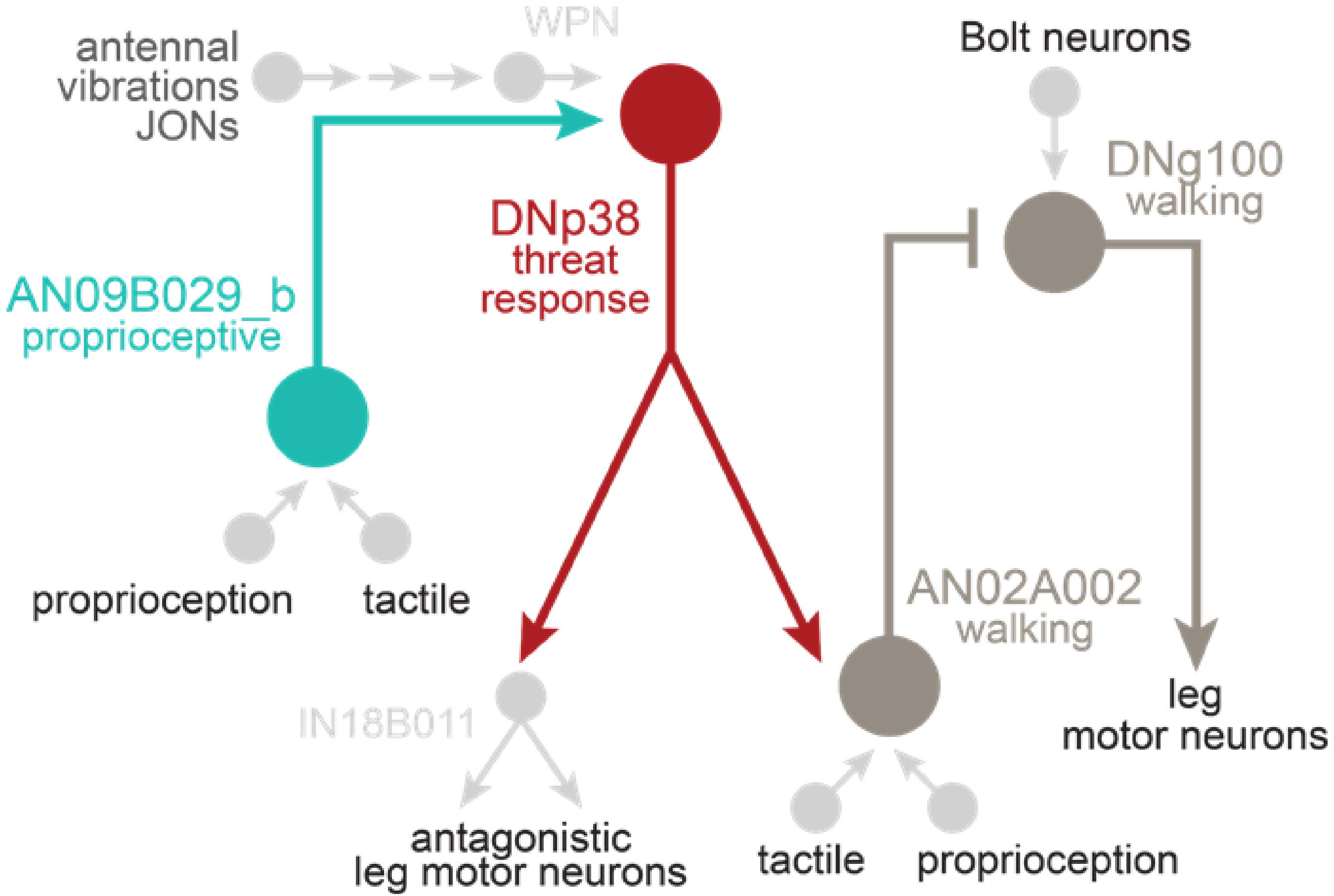


influence to:
wing motor
(power)

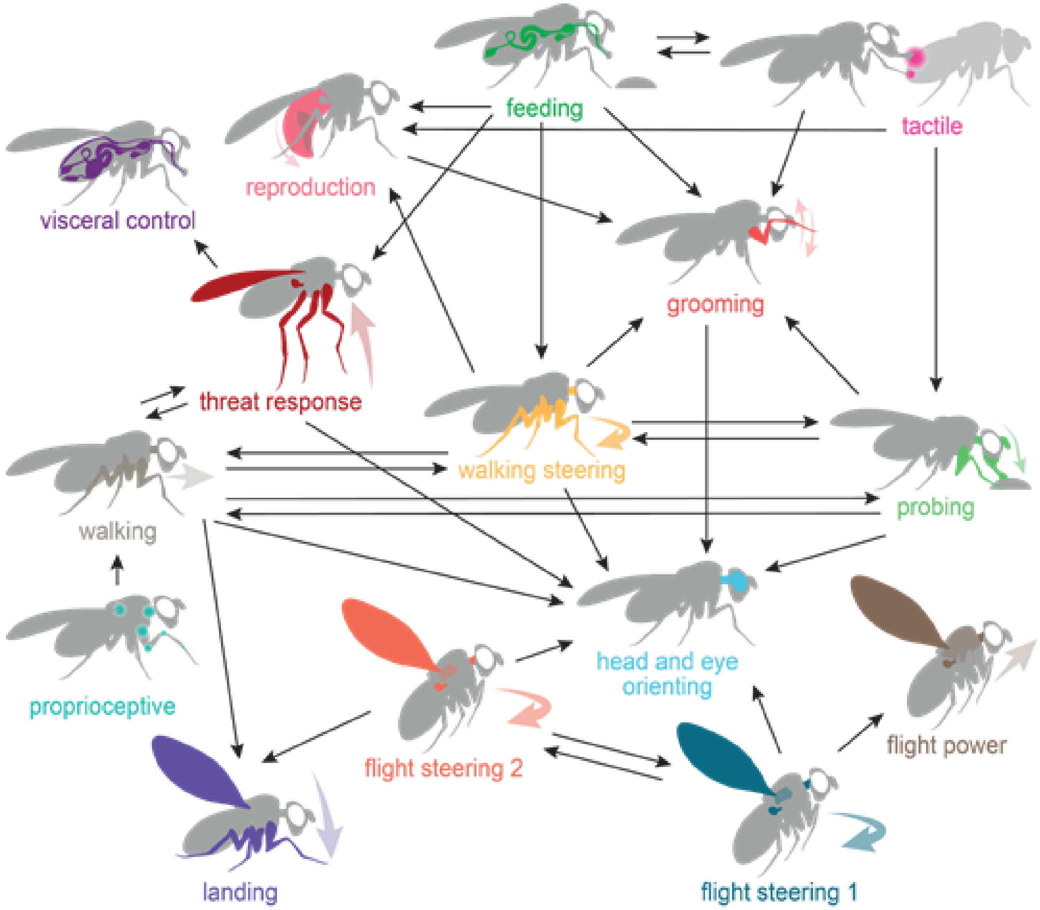
ascending and descending neuron superclusters associate with specific behaviors



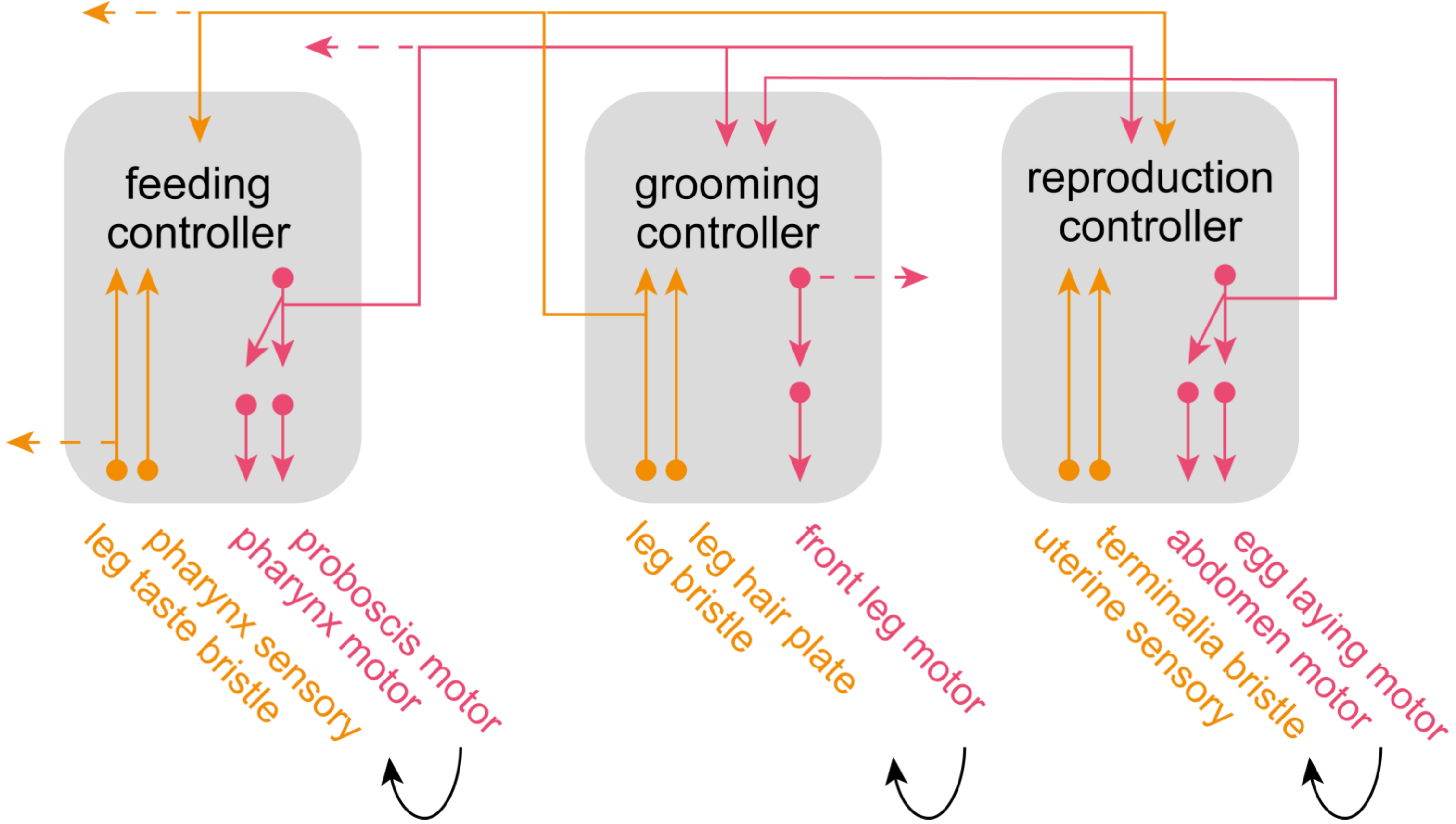
ascending and descending neuron superclusters influence each other



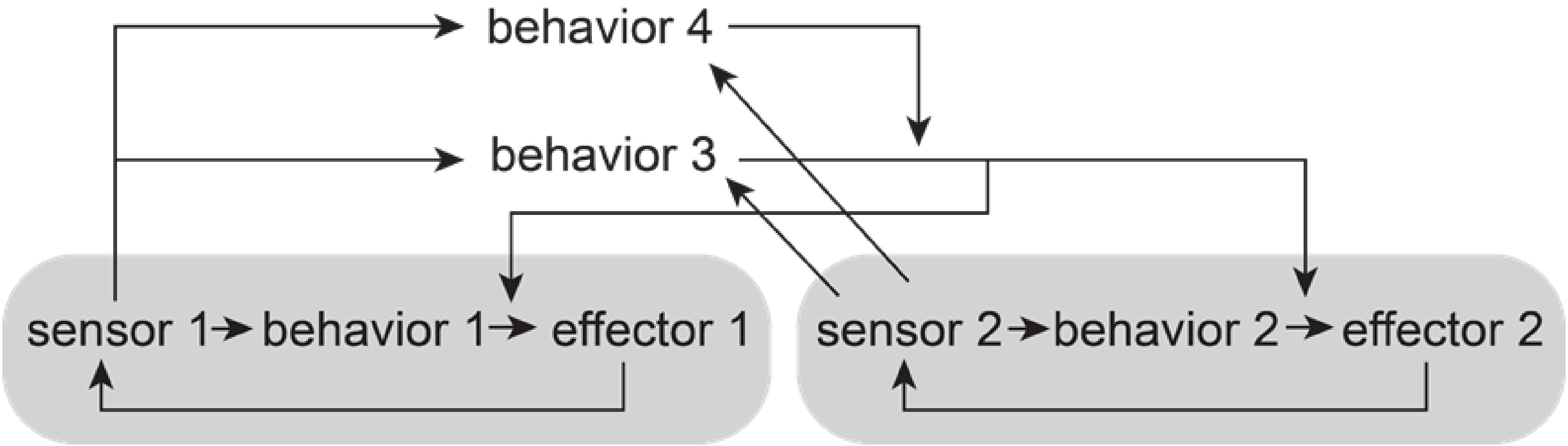
ascending and descending neuron superclusters influence each other



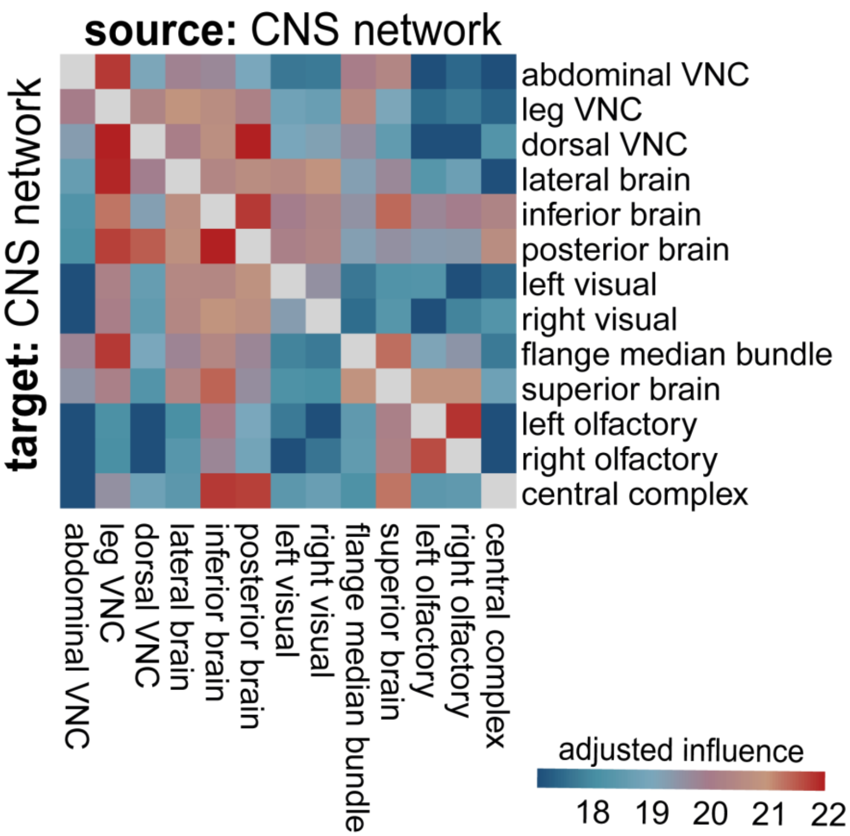
low-level sensor-effector units are chained by long-ranged connections



the arrangement of ascending and descending superclusters is analogous to a **subsumption architecture**



classical "central controllers" are not very influential



Rachel
Wilson

Jan
Drugowitsch

Helen
Yang

Zaki
Ajabi

+ Sophia Renauld, Mohammed Abdal Monium Osman,
Matthew F. Collie, Jingxuan Fan, Diego A. Pacheco

the BANC is ready for use

primary data visualization

neuroglancer Public

http://ng.banc.community/view

(meta)data browsing

https://codex.flywire.ai/?dataset=banc

data archives

HARVARD Dataverse

github.com/htem/BANC-project
https://doi.org/10.7910/DVN/8TFGGB

tools

github.com/flyconnectome/bancr
github.com/CAVEconnectome/

acknowledgements

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